

# Birds Classification based on Bird Sound

**Guided by :**

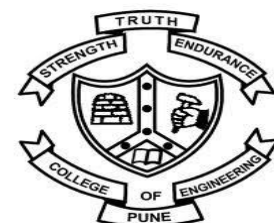
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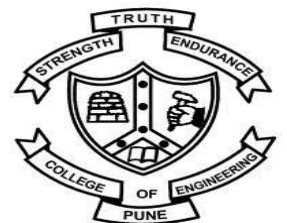
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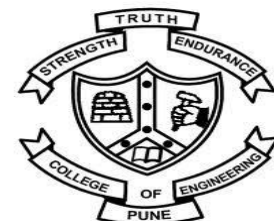
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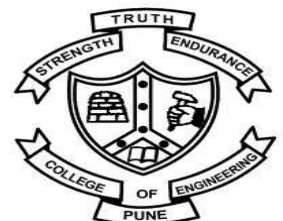
## Problem Statement

- To propose a system that aims to automatically recognize bird species or even individual birds by their sound in field conditions.



# Introduction

- Birds can be heard even in big cities.
- Most common species by their sound and experts can recognize hundreds of species only by their sound.
- Recognition of bird species by their sounds is a typical pattern recognition problem.



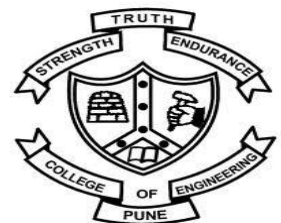
# Literature Survey:

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| TITLE                                                                               | CONCLUSION                                                                                                                                              |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Speaker recognition using mel-frequency cepstrum coefficients and sum square error  | The use of Mel Frequency Cepstral Coefficients with Vector Quantization for speaker verification. VQ is used to produce a vector of each speech signal. |
| Investigation of Different CNN-Based Models for Improved Bird Sound Classification, | Different types of time-frequency representations called Mel, harmonic and percussive spectrogram. CNN fused model gives balanced accuracy of 86.31 %.  |

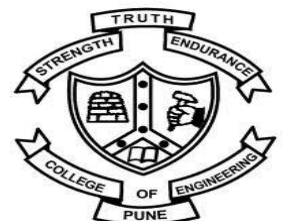
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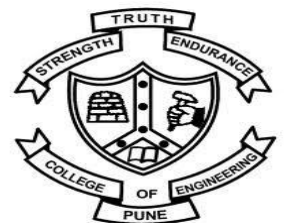
| TITLE                                                                                     | CONCLUSION                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bird sound recognition using a convolutional neural network                               | Developed a methodology to classify species using spectrogram. A pre-trained MobileNet CNN model was used to differentiate species. STFT of each audio file given to input to CNN model. This experiment shows the color map of the spectrogram gives more advantages than a normal spectrogram. |
| Bird species recognition using unsupervised modeling of individual vocalization elements. | Suggested tonal-based features to identify tonal bird sounds. The Gaussian mixture model is used to model 95 bird species. Each bird sound contains white noise and environmental noise. The Tonal-based approach provides better performance compared to Mel Frequency Cepstral Coefficients.   |

| TITLE                                                                       | CONCLUSION                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Analysis of LFCC Feature Extraction in Baby Crying Classification using KNN | Designed a new feature extraction technique called Linear Frequency Cepstral Coefficients (LFCC). LFCC uses a Linear Filter bank K-nearest neighbour algorithm is implemented to detect baby is crying or not.                                                                            |
| Acoustic Classification of Bird Species.                                    | This paper has proposed a method identifying birds by their sounds using Mel Frequency Cepstral Coefficients (MFCC) and Gammatone Frequency Cepstral Coefficients (GFCC) for classification. The feature set is evaluated using the Support Vector Machine and Artificial Neural Network. |



# Aim and Objective

- To create a dataset of bird sound.
- To study and Extract features from bird sound.
- To select optimum feature for bird classification.
- To perform classification of bird species using different machine learning algorithms.
- To compare the performance of various classification algorithms.



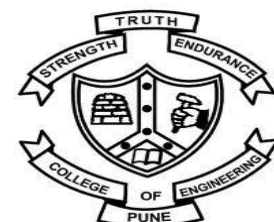


# Dataset

| Sr. No. | Bird species                 | Scientific name                     | Number of Files   |
|---------|------------------------------|-------------------------------------|-------------------|
| 1       | Red-whiskered Bulbul(RB)     | Pycnonotus jocosus                  | 236               |
| 2       | Great Tilt(GT)               | Parus major                         | 243               |
| 3       | Willow Warbler(WW)           | Phylloscopus trochilus              | 262               |
| 4       | European Pied Flycatcher(EF) | Ficedula hypoleuca                  | 264               |
| 5       | Common Blackbird(CB)         | Turdus merula                       | 256               |
| 6       | Common Redstart(CR)          | Phoenicurus phoenicurus             | 188               |
| 7       | Greenish Warbler(GW)         | phylloscopus torchiloides viridanus | 207               |
| 8       | Blue Jay(BJ)                 | Cyanocitta cristata                 | 276               |
| 9       | Common Snipe(CS)             | Gallinago gallinago                 | 244               |
| 10      | Whistling Duck(WD)           | Denfrocygna viduta                  | 182               |
|         |                              |                                     | <b>Total 2358</b> |

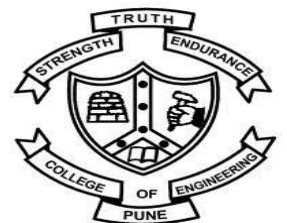
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# Feature Extraction

- 1) Mel – Frequency Cepstral Coefficients (MFCC)
- 2) Linear Frequency Cepstral Coefficient (LFCC)
- 3) Gammatone frequency cepstral coefficient (GFCC)



# Feature Extraction Block Diagram

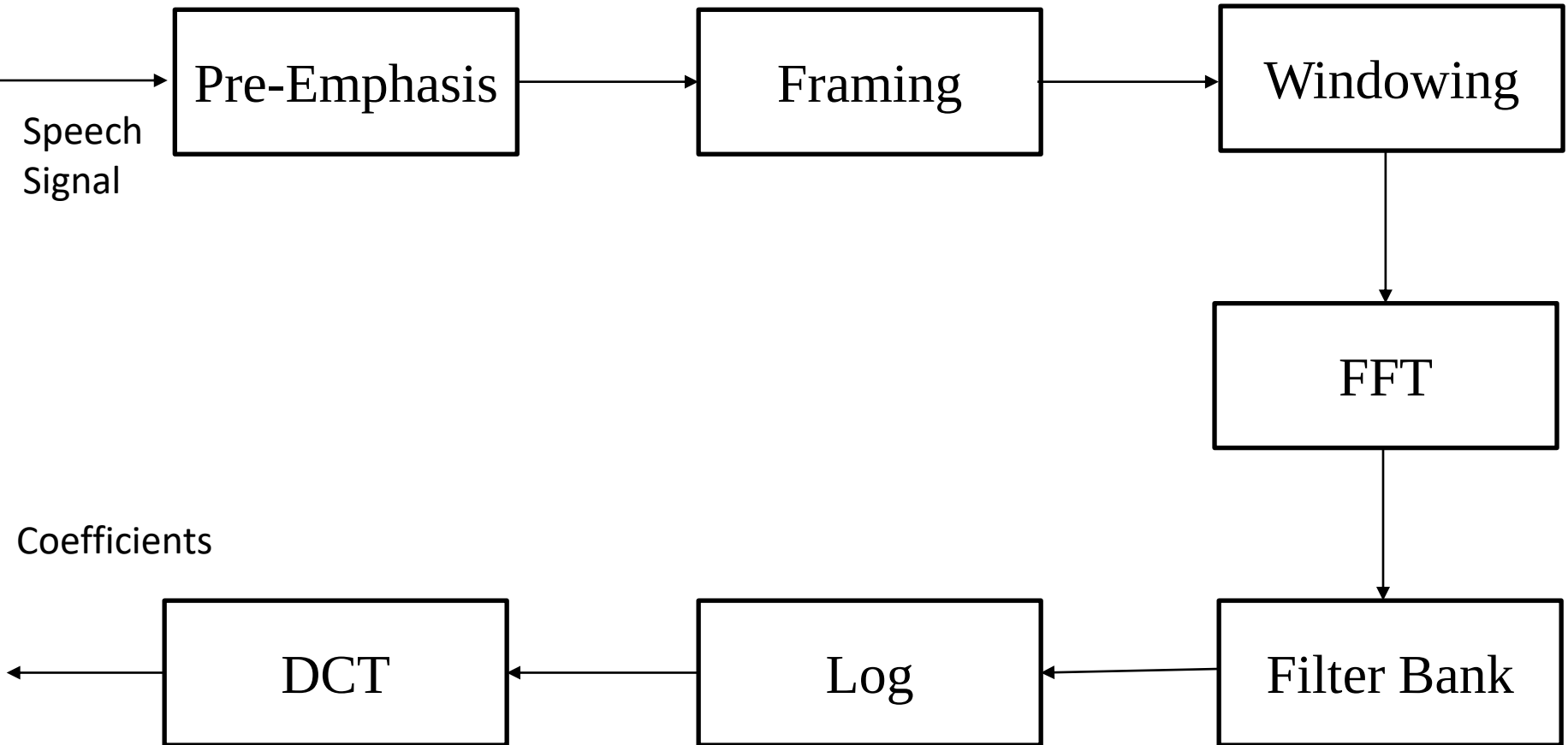
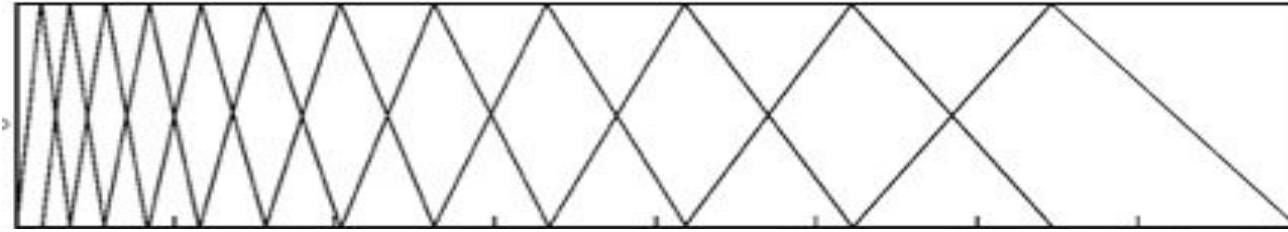
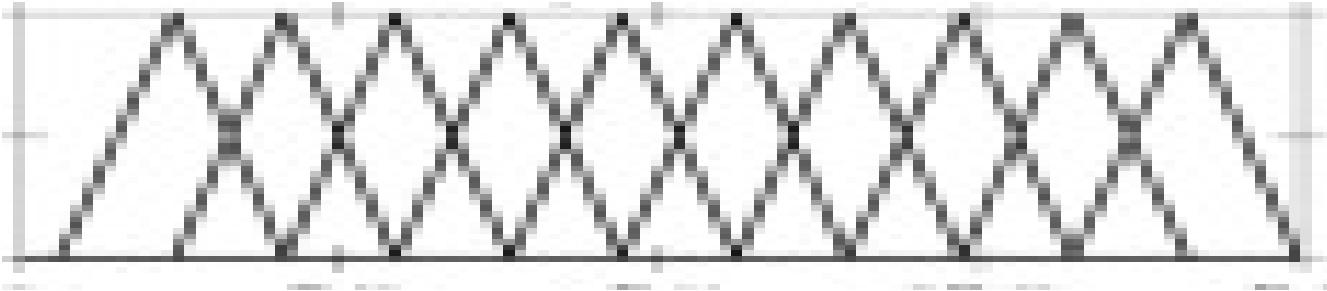


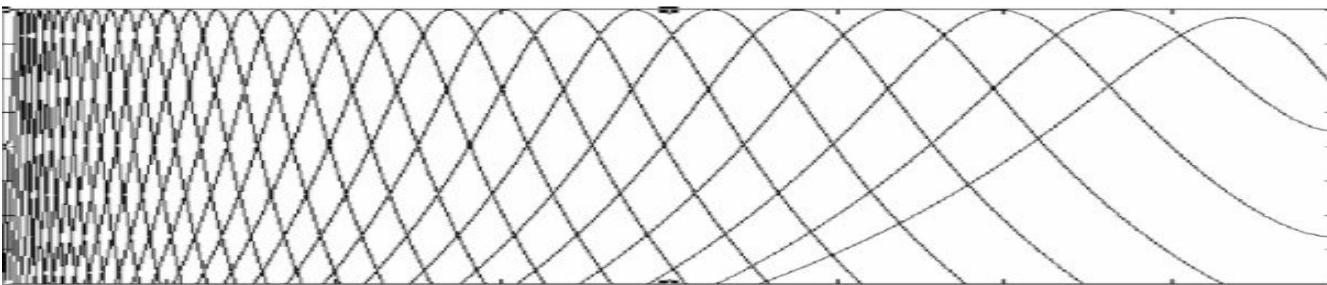
Fig. 1: Block Diagram of Feature Extraction



Mel Filter Bank



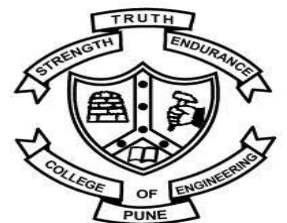
Linear Filter Bank



Gammatone Filter Bank

# Result of Feature Extraction

- This section discusses the result of all feature vectors of MFCC, LFCC and GFCC.
- The experiment is performed on 13,26 and 39 coefficients.
- The best result is shown in **Green** color and the worst result shown in **Red** color.



# Support Vector Machine

- The main goal of the support vector machine is to find a hyperplane between two classes.
- To separate two different classes, the margin between two classes is maximum.

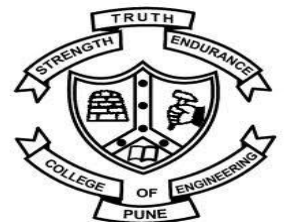


Table 1: Recognition of result with MFCC (13) features

|                          | Number of correctly recognized bird sound in the class |    |    |    |    |    |    |    |    |    | RSA |
|--------------------------|--------------------------------------------------------|----|----|----|----|----|----|----|----|----|-----|
|                          | RB                                                     | GT | WW | EF | CB | CR | GW | BJ | CS | WD |     |
| RB                       | 36                                                     | 1  | 2  | 0  | 0  | 0  | 0  | 1  | 4  | 0  | 82  |
| GT                       | 3                                                      | 45 | 0  | 1  | 0  | 0  | 2  | 0  | 0  | 0  | 88  |
| WW                       | 2                                                      | 0  | 37 | 4  | 0  | 0  | 1  | 4  | 0  | 0  | 77  |
| EF                       | 0                                                      | 0  | 9  | 47 | 1  | 0  | 0  | 1  | 0  | 0  | 81  |
| CB                       | 0                                                      | 0  | 2  | 6  | 43 | 2  | 0  | 0  | 0  | 0  | 81  |
| CR                       | 1                                                      | 0  | 5  | 0  | 0  | 59 | 0  | 0  | 0  | 0  | 76  |
| GW                       | 4                                                      | 1  | 3  | 0  | 0  | 0  | 48 | 1  | 0  | 0  | 84  |
| BJ                       | 0                                                      | 0  | 0  | 0  | 0  | 0  | 4  | 43 | 8  | 6  | 70  |
| CS                       | 1                                                      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 39 | 1  | 95  |
| WD                       | 0                                                      | 0  | 0  | 0  | 0  | 0  | 0  | 2  | 0  | 36 | 92  |
| Rel.                     | 75                                                     | 96 | 64 | 81 | 98 | 90 | 87 | 83 | 76 | 84 |     |
| Overall Accuracy: 82.38% |                                                        |    |    |    |    |    |    |    |    |    |     |

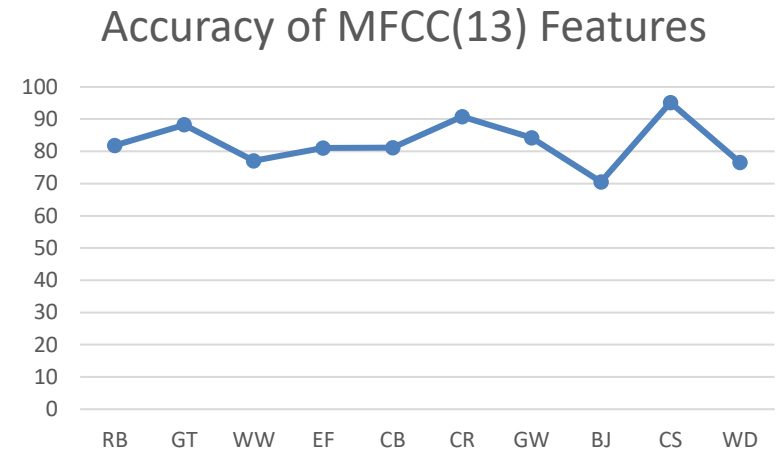
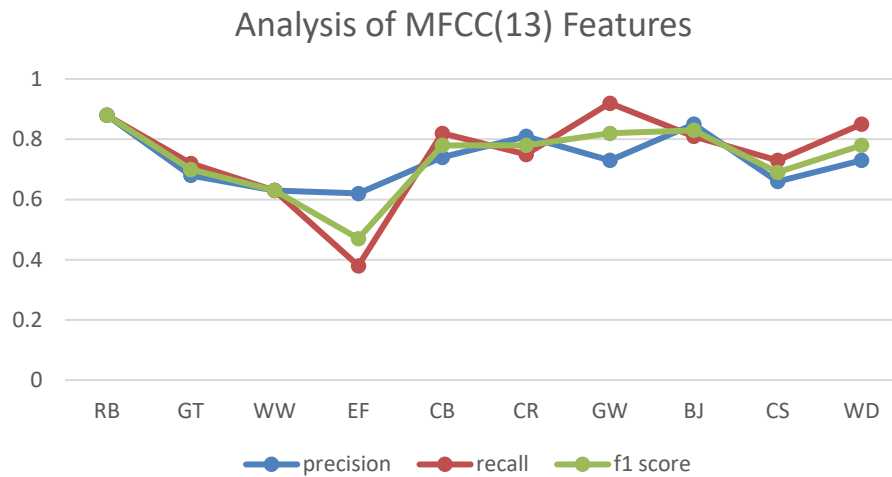


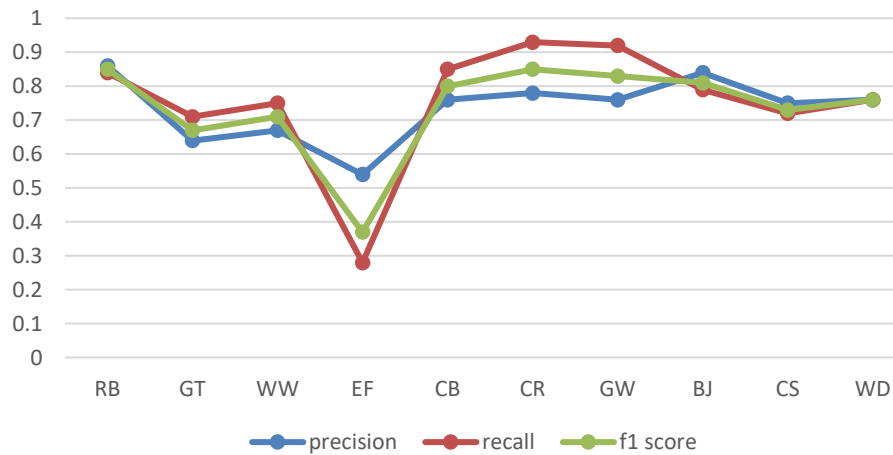
Fig 2 : Analysis and Accuracy of MFCC (13) Features



Table 2 : Recognition of result with MFCC (26) features

|                          | Number of correctly recognized bird sound in the class |    |    |    |    |    |    |    |    |    | % RSA |
|--------------------------|--------------------------------------------------------|----|----|----|----|----|----|----|----|----|-------|
|                          | RB                                                     | GT | WW | EF | CB | CR | GW | BJ | CS | WD |       |
| RB                       | 57                                                     | 1  | 0  | 0  | 0  | 0  | 0  | 5  | 2  | 3  | 85    |
| GT                       | 0                                                      | 55 | 0  | 13 | 3  | 2  | 4  | 0  | 1  | 0  | 67    |
| WW                       | 6                                                      | 1  | 58 | 2  | 0  | 3  | 3  | 0  | 1  | 3  | 71    |
| EF                       | 0                                                      | 21 | 24 | 25 | 4  | 7  | 6  | 0  | 0  | 2  | 37    |
| CB                       | 0                                                      | 3  | 0  | 2  | 67 | 3  | 0  | 0  | 4  | 0  | 80    |
| CR                       | 0                                                      | 1  | 0  | 2  | 0  | 63 | 0  | 2  | 0  | 0  | 85    |
| GW                       | 2                                                      | 0  | 1  | 0  | 2  | 0  | 54 | 0  | 0  | 0  | 83    |
| BJ                       | 1                                                      | 0  | 2  | 0  | 1  | 2  | 4  | 61 | 6  | 0  | 81    |
| CS                       | 0                                                      | 3  | 2  | 0  | 11 | 0  | 0  | 0  | 48 | 3  | 73    |
| WD                       | 0                                                      | 1  | 0  | 2  | 0  | 1  | 0  | 5  | 2  | 35 | 76    |
| % Rel.                   | 86                                                     | 64 | 67 | 54 | 76 | 78 | 76 | 84 | 75 | 76 |       |
| Overall Accuracy: 81.33% |                                                        |    |    |    |    |    |    |    |    |    |       |

Analysis of MFCC(26) Features



Accuracy of MFCC(26) Features

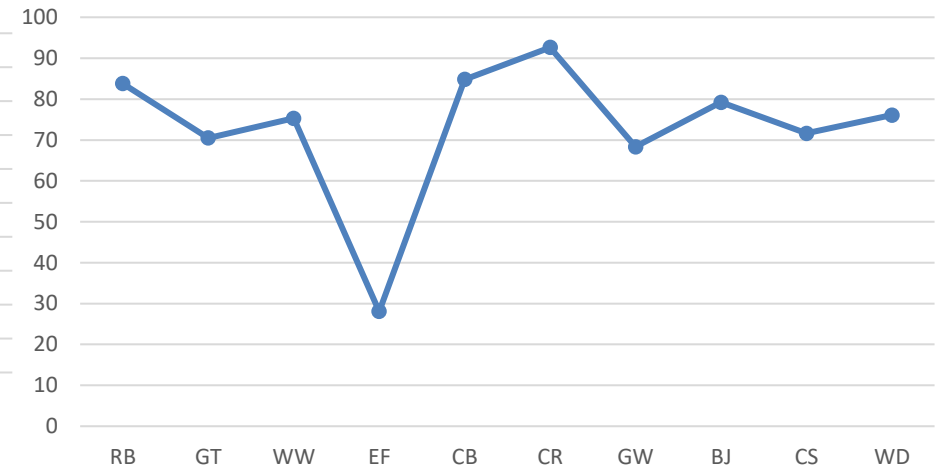


Fig 3 : Analysis and Accuracy of MFCC (26) Features

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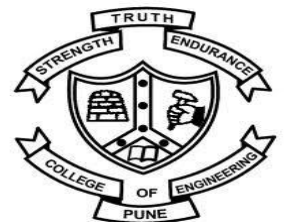
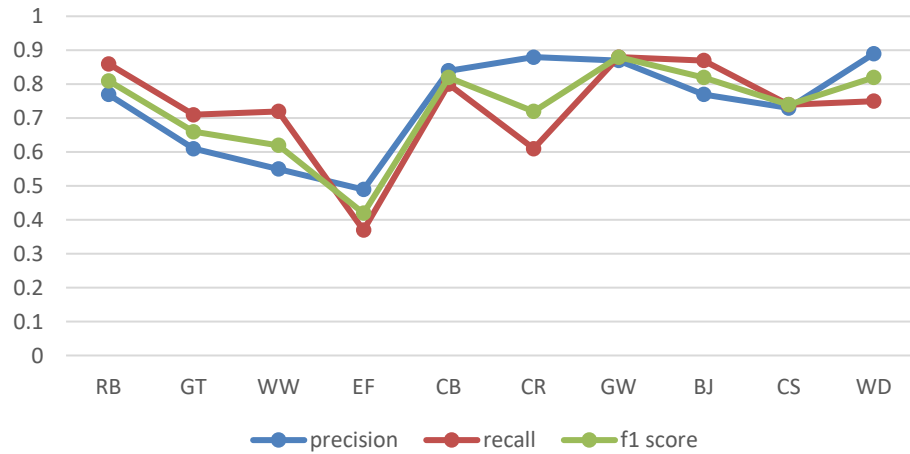
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Table 3 : Recognition of result with MFCC (39) features

|                          | Number of correctly recognized bird sound |    |    |    |    |    |    |    |    |    | RSA |
|--------------------------|-------------------------------------------|----|----|----|----|----|----|----|----|----|-----|
|                          | RB                                        | GT | WW | EF | CB | CR | GW | BJ | CS | WD |     |
| RB                       | 48                                        | 1  | 3  | 0  | 0  | 0  | 0  | 1  | 3  | 0  | 81  |
| GT                       | 0                                         | 45 | 2  | 9  | 2  | 2  | 1  | 0  | 2  | 0  | 66  |
| WW                       | 4                                         | 2  | 53 | 7  | 1  | 0  | 3  | 4  | 0  | 0  | 62  |
| EF                       | 1                                         | 20 | 25 | 31 | 4  | 1  | 1  | 0  | 0  | 1  | 42  |
| CB                       | 1                                         | 1  | 1  | 2  | 65 | 0  | 2  | 1  | 8  | 0  | 82  |
| CR                       | 3                                         | 2  | 7  | 5  | 0  | 37 | 0  | 4  | 0  | 3  | 72  |
| GW                       | 0                                         | 0  | 4  | 2  | 1  | 0  | 61 | 1  | 0  | 0  | 88  |
| BJ                       | 3                                         | 0  | 0  | 0  | 1  | 1  | 2  | 78 | 5  | 0  | 82  |
| CS                       | 2                                         | 0  | 1  | 2  | 2  | 0  | 0  | 11 | 55 | 1  | 74  |
| WD                       | 0                                         | 3  | 1  | 5  | 1  | 1  | 0  | 1  | 2  | 42 | 82  |
| Rel.                     | 77                                        | 61 | 55 | 49 | 84 | 88 | 87 | 77 | 73 | 89 |     |
| Overall Accuracy: 82.21% |                                           |    |    |    |    |    |    |    |    |    |     |

Analysis of MFCC(39) Features



Accuracy of MFCC(39) Features

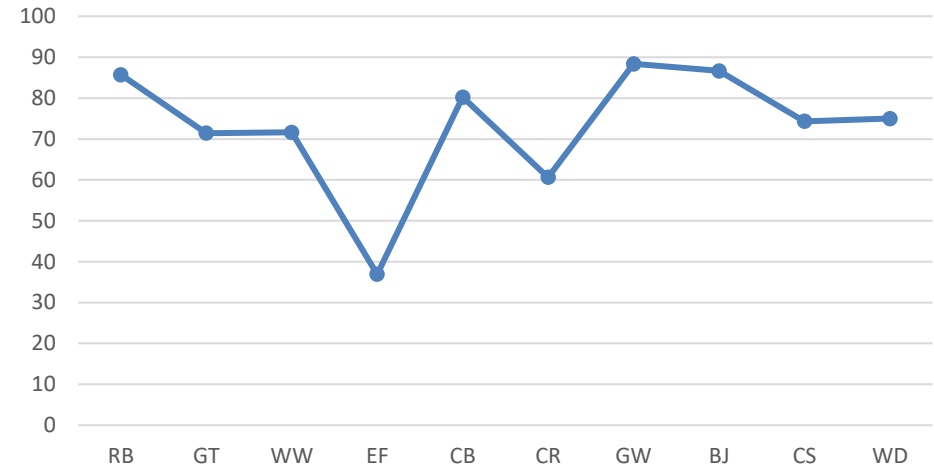
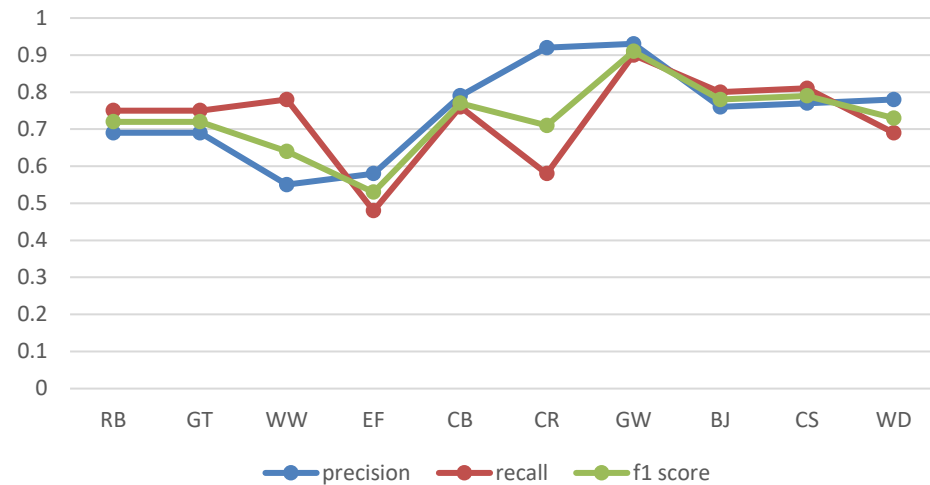


Fig 4 : Analysis and Accuracy of MFCC (39) Features

Table 4 : Recognition of result with LFCC (13) features

|                           | Number of correctly recognized bird sound |    |    |    |    |    |    |    |    |    | Acc % |
|---------------------------|-------------------------------------------|----|----|----|----|----|----|----|----|----|-------|
|                           | RB                                        | GT | WW | EF | CB | CR | GW | BJ | CS | WD |       |
| RB                        | 50                                        | 0  | 1  | 0  | 0  | 0  | 0  | 2  | 3  | 0  | 89    |
| GT                        | 1                                         | 38 | 0  | 1  | 2  | 0  | 0  | 0  | 0  | 0  | 90    |
| WW                        | 0                                         | 10 | 37 | 9  | 0  | 1  | 2  | 0  | 0  | 3  | 71    |
| EF                        | 0                                         | 2  | 1  | 53 | 0  | 0  | 0  | 0  | 0  | 0  | 95    |
| CB                        | 0                                         | 0  | 1  | 5  | 42 | 0  | 1  | 0  | 0  | 0  | 86    |
| CR                        | 0                                         | 0  | 3  | 2  | 0  | 55 | 0  | 2  | 0  | 0  | 77    |
| GW                        | 0                                         | 0  | 3  | 0  | 0  | 0  | 57 | 1  | 0  | 0  | 93    |
| BJ                        | 2                                         | 0  | 0  | 0  | 0  | 0  | 4  | 39 | 10 | 3  | 80    |
| CS                        | 4                                         | 0  | 0  | 0  | 0  | 0  | 0  | 8  | 60 | 0  | 74    |
| WD                        | 0                                         | 0  | 0  | 0  | 1  | 0  | 0  | 0  | 0  | 34 | 97    |
| % Rel                     | 88                                        | 95 | 80 | 76 | 93 | 96 | 89 | 75 | 89 | 85 |       |
| Overall Accuracy: 85.53 % |                                           |    |    |    |    |    |    |    |    |    |       |

Analysis of LFCC(13) Features



Accuracy of LFCC(13) Features

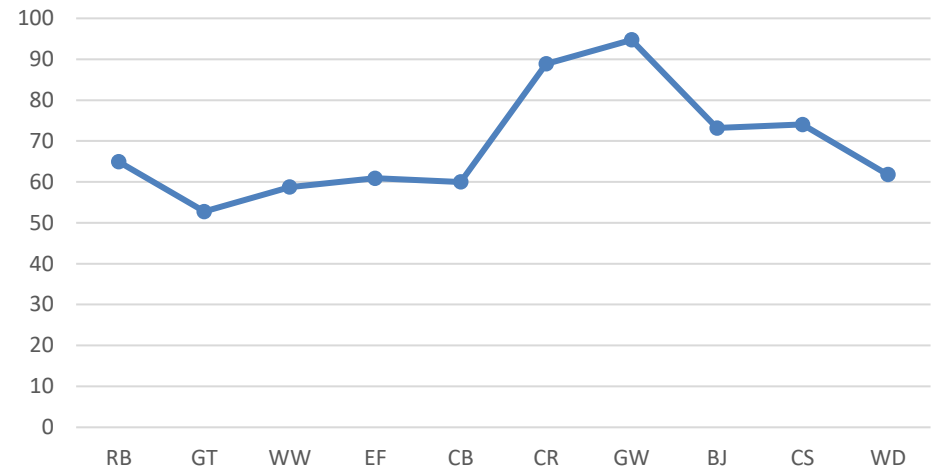


Fig 5 : Analysis and Accuracy of LFCC (13) Features

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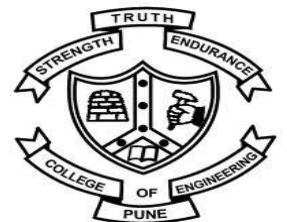
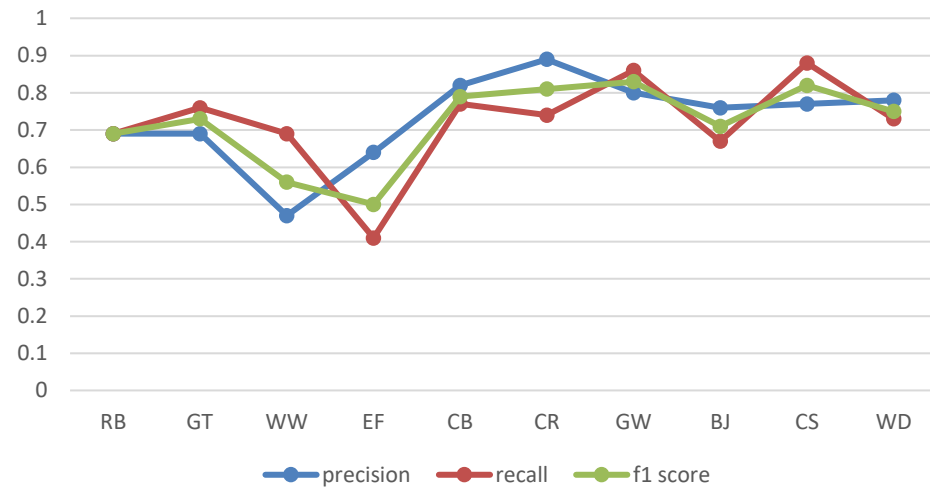
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Table 5 : Recognition of result with LFCC (26) features

|                          | Number of correctly recognized bird sound in the class |    |    |    |    |    |    |    |    |    | RSA |
|--------------------------|--------------------------------------------------------|----|----|----|----|----|----|----|----|----|-----|
|                          | RB                                                     | GT | WW | EF | CB | CR | GW | BJ | CS | WD |     |
| RB                       | 44                                                     | 1  | 9  | 0  | 0  | 1  | 0  | 6  | 3  | 0  | 69  |
| GT                       | 1                                                      | 52 | 6  | 3  | 3  | 0  | 2  | 0  | 1  | 0  | 73  |
| WW                       | 4                                                      | 1  | 53 | 8  | 0  | 0  | 2  | 6  | 2  | 1  | 56  |
| EF                       | 0                                                      | 13 | 25 | 36 | 5  | 3  | 0  | 0  | 0  | 5  | 50  |
| CB                       | 3                                                      | 2  | 1  | 5  | 59 | 0  | 3  | 1  | 3  | 0  | 79  |
| CR                       | 2                                                      | 3  | 8  | 3  | 0  | 48 | 0  | 1  | 0  | 0  | 81  |
| GW                       | 2                                                      | 2  | 3  | 0  | 0  | 0  | 49 | 1  | 0  | 0  | 83  |
| BJ                       | 2                                                      | 0  | 7  | 0  | 1  | 0  | 5  | 56 | 8  | 4  | 71  |
| CS                       | 5                                                      | 1  | 1  | 0  | 1  | 0  | 0  | 2  | 72 | 0  | 82  |
| WD                       | 1                                                      | 0  | 0  | 1  | 3  | 2  | 0  | 1  | 5  | 35 | 75  |
| Rel.                     | 69                                                     | 69 | 47 | 64 | 82 | 89 | 80 | 76 | 77 | 78 |     |
| Overall Accuracy: 85.49% |                                                        |    |    |    |    |    |    |    |    |    |     |

Analysis of LFCC(26) Features



Accuracy of LFCC(26) Features

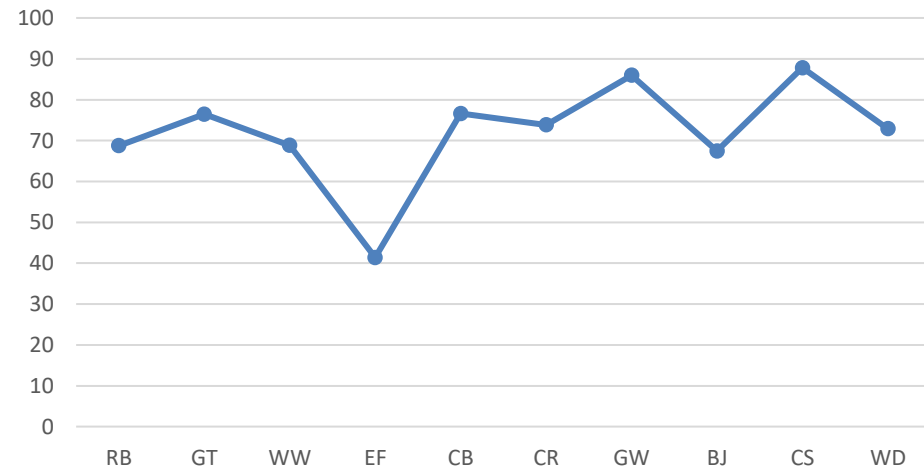


Fig 6 : Analysis and Accuracy of LFCC (26) Features

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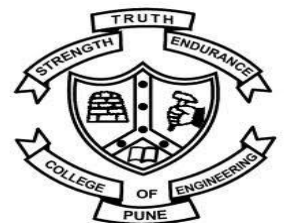
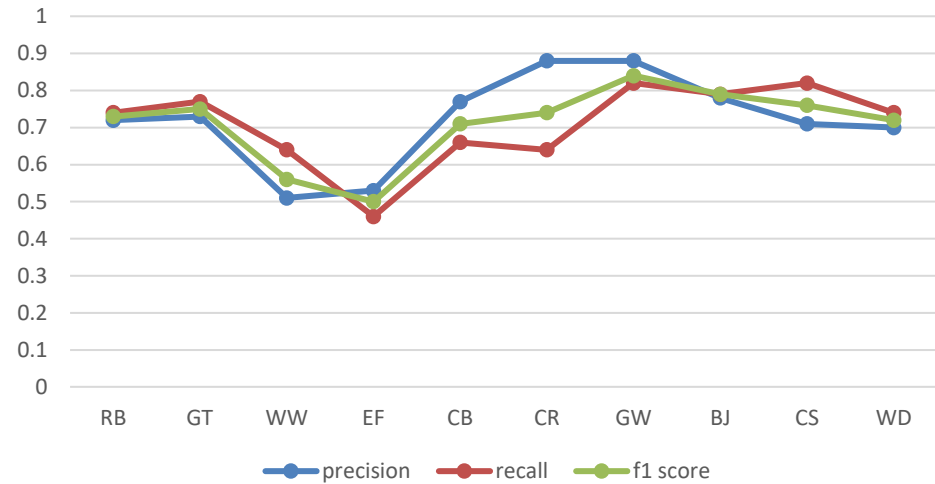
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Table 6: Recognition of result with LFCC (39) features

|                          | Number of correctly recognized bird sound in the class |    |    |    |    |    |    |    |    |    | RSA |
|--------------------------|--------------------------------------------------------|----|----|----|----|----|----|----|----|----|-----|
|                          | RB                                                     | GT | WW | EF | CB | CR | GW | BJ | CS | WD |     |
| RB                       | 49                                                     | 0  | 7  | 1  | 1  | 1  | 0  | 3  | 2  | 2  | 73  |
| GT                       | 1                                                      | 56 | 1  | 5  | 2  | 0  | 2  | 4  | 2  | 0  | 75  |
| WW                       | 1                                                      | 2  | 46 | 13 | 0  | 0  | 1  | 1  | 1  | 3  | 56  |
| EF                       | 0                                                      | 10 | 14 | 39 | 7  | 2  | 0  | 0  | 0  | 12 | 50  |
| CB                       | 2                                                      | 2  | 2  | 9  | 51 | 0  | 1  | 3  | 7  | 0  | 71  |
| CR                       | 0                                                      | 2  | 10 | 4  | 0  | 35 | 0  | 0  | 0  | 0  | 84  |
| GW                       | 1                                                      | 4  | 4  | 1  | 0  | 0  | 49 | 1  | 0  | 0  | 84  |
| BJ                       | 3                                                      | 0  | 2  | 0  | 1  | 0  | 3  | 68 | 8  | 1  | 79  |
| CS                       | 5                                                      | 0  | 0  | 0  | 1  | 0  | 0  | 6  | 64 | 0  | 86  |
| WD                       | 0                                                      | 1  | 1  | 1  | 3  | 2  | 0  | 1  | 6  | 42 | 72  |
| Rel.                     | 72                                                     | 73 | 55 | 53 | 77 | 88 | 88 | 78 | 71 | 70 |     |
| Overall Accuracy: 85.88% |                                                        |    |    |    |    |    |    |    |    |    |     |

Analysis of LFCC(39) Features



Accuracy of LFCC(39) Features

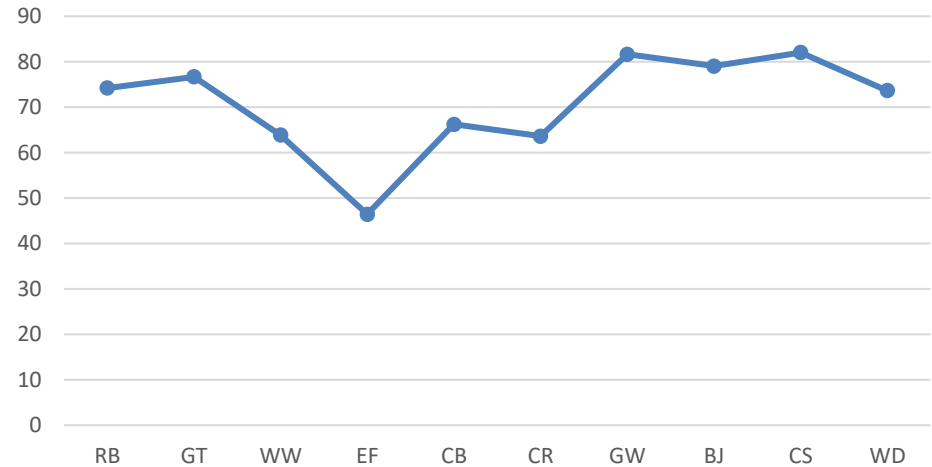


Fig 7 : Analysis and Accuracy of LFCC (39) Features

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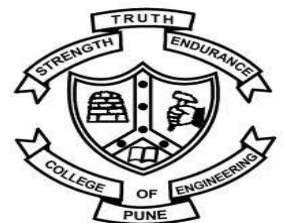
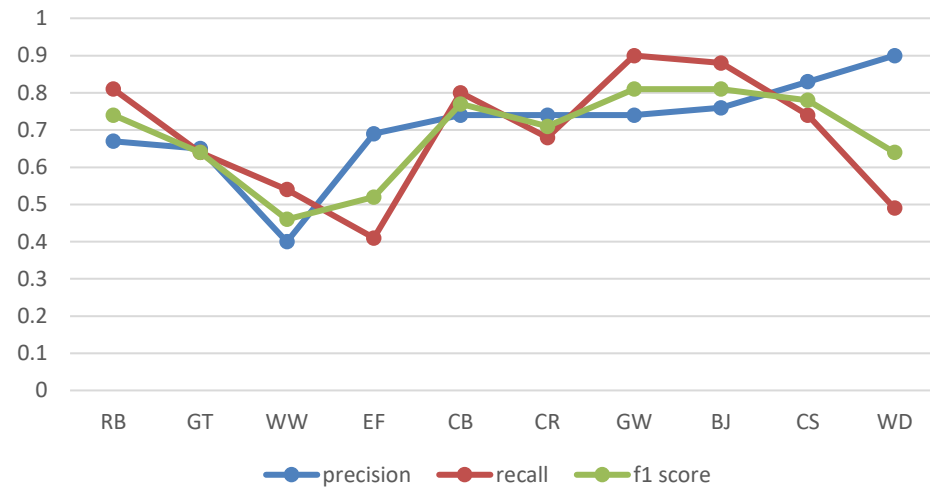
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Table 7: Recognition of result with GFCC (13) features

|                          | Number of correctly recognized bird sound |    |    |    |    |    |    |    |    |    | RSA |
|--------------------------|-------------------------------------------|----|----|----|----|----|----|----|----|----|-----|
|                          | RB                                        | GT | WW | EF | CB | CR | GW | BJ | CS | WD |     |
| RB                       | 56                                        | 0  | 4  | 1  | 1  | 0  | 0  | 3  | 4  | 0  | 74  |
| GT                       | 1                                         | 46 | 4  | 7  | 11 | 2  | 0  | 0  | 0  | 1  | 64  |
| WW                       | 17                                        | 4  | 39 | 2  | 1  | 2  | 5  | 2  | 0  | 0  | 46  |
| EF                       | 0                                         | 13 | 27 | 34 | 1  | 0  | 6  | 0  | 0  | 1  | 52  |
| CB                       | 1                                         | 4  | 0  | 1  | 64 | 0  | 7  | 0  | 3  | 0  | 77  |
| CR                       | 0                                         | 2  | 7  | 1  | 0  | 34 | 1  | 4  | 0  | 1  | 71  |
| GW                       | 0                                         | 0  | 2  | 3  | 1  | 0  | 54 | 0  | 0  | 0  | 81  |
| BJ                       | 3                                         | 0  | 2  | 0  | 0  | 2  | 0  | 77 | 4  | 0  | 81  |
| CS                       | 5                                         | 0  | 3  | 0  | 5  | 0  | 0  | 7  | 58 | 0  | 78  |
| WD                       | 0                                         | 2  | 10 | 0  | 2  | 6  | 0  | 8  | 1  | 28 | 64  |
| Rel.                     | 67                                        | 65 | 40 | 69 | 74 | 74 | 74 | 76 | 83 | 90 |     |
| Overall Accuracy: 87.64% |                                           |    |    |    |    |    |    |    |    |    |     |

Analysis of GFCC(13) Features



Accuracy of GFCC(13) Features

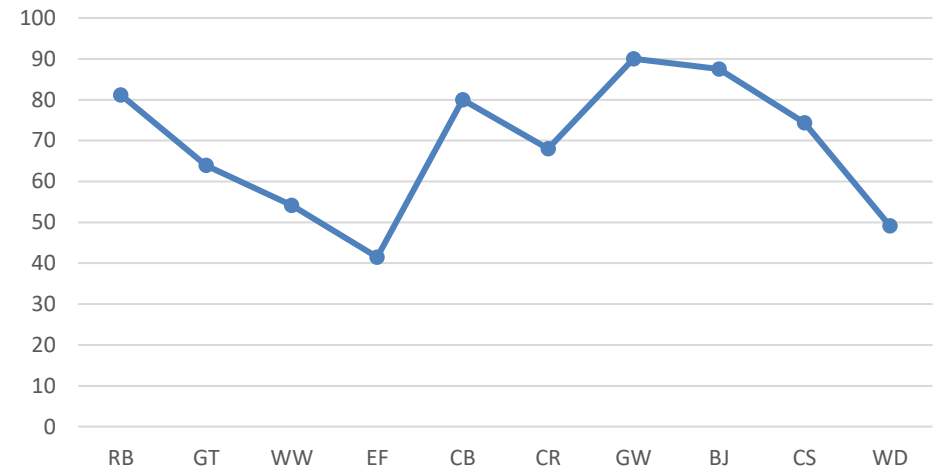
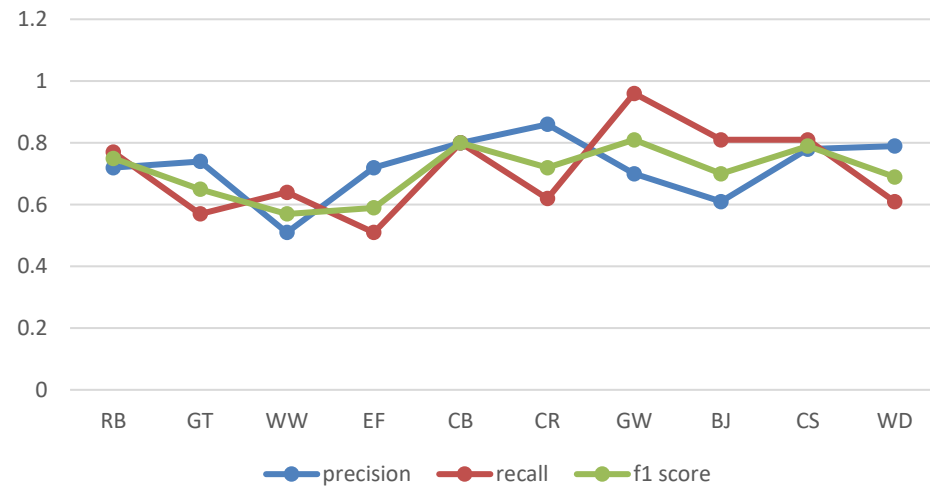


Fig 8 : Analysis and Accuracy of GFCC (13) Features

Table 8: Recognition of result with GFCC (26) features

|                          | Number of correctly recognized bird sound |    |    |    |    |    |    |    |    |    | RSA |
|--------------------------|-------------------------------------------|----|----|----|----|----|----|----|----|----|-----|
|                          | RB                                        | GT | WW | EF | CB | CR | GW | BJ | CS | WD |     |
| RB                       | 58                                        | 0  | 2  | 0  | 0  | 0  | 0  | 9  | 6  | 0  | 75  |
| GT                       | 0                                         | 40 | 6  | 10 | 11 | 0  | 0  | 0  | 0  | 3  | 65  |
| WW                       | 10                                        | 5  | 51 | 0  | 1  | 2  | 6  | 5  | 0  | 0  | 57  |
| EF                       | 0                                         | 3  | 32 | 46 | 0  | 1  | 6  | 0  | 0  | 3  | 59  |
| CB                       | 1                                         | 2  | 0  | 3  | 66 | 0  | 5  | 2  | 4  | 0  | 80  |
| CR                       | 1                                         | 0  | 4  | 5  | 0  | 36 | 1  | 8  | 0  | 3  | 72  |
| GW                       | 0                                         | 0  | 1  | 0  | 0  | 0  | 47 | 1  | 0  | 0  | 81  |
| BJ                       | 4                                         | 0  | 0  | 0  | 1  | 1  | 1  | 61 | 7  | 0  | 70  |
| CS                       | 5                                         | 0  | 2  | 0  | 2  | 0  | 0  | 5  | 59 | 0  | 79  |
| WD                       | 1                                         | 4  | 2  | 0  | 2  | 2  | 1  | 9  | 0  | 33 | 69  |
| Rel.                     | 72                                        | 74 | 51 | 72 | 80 | 86 | 70 | 61 | 78 | 79 |     |
| Overall Accuracy: 87.44% |                                           |    |    |    |    |    |    |    |    |    |     |

Analysis of GFCC(26) Features



Accuracy of GFCC(26) Features

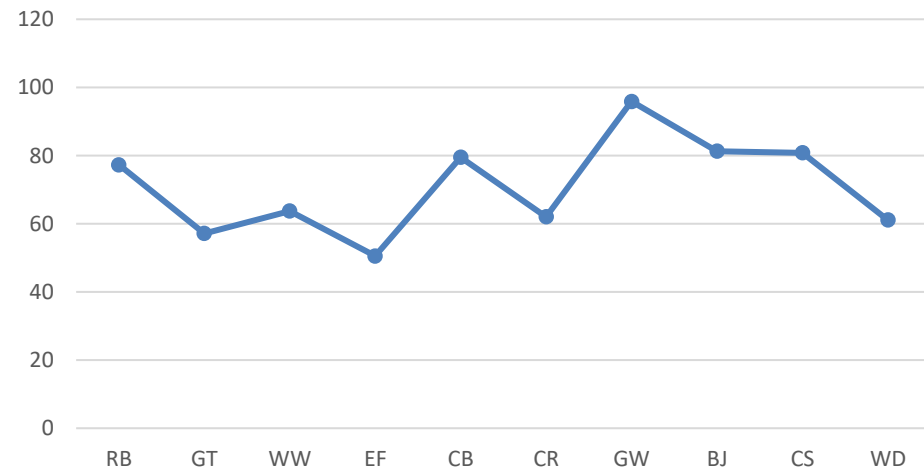
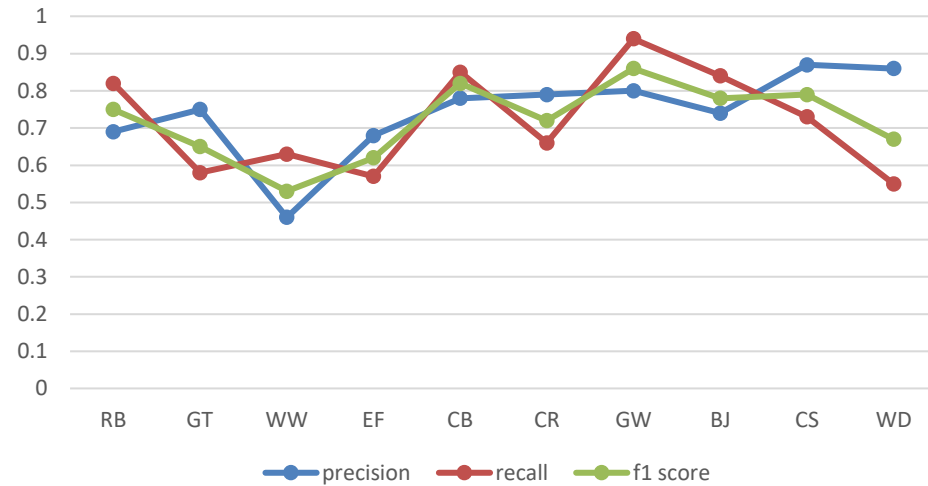


Fig 9 : Analysis and Accuracy of GFCC (26) Features

Table 9: Recognition of result with GFCC (39) features

|                          | Number of correctly recognized bird sound |    |    |    |    |    |    |    |    |    | RSA |
|--------------------------|-------------------------------------------|----|----|----|----|----|----|----|----|----|-----|
|                          | RB                                        | GT | WW | EF | CB | CR | GW | BJ | CS | WD |     |
| RB                       | 46                                        | 0  | 2  | 0  | 0  | 0  | 0  | 5  | 3  | 0  | 75  |
| GT                       | 0                                         | 41 | 4  | 14 | 9  | 1  | 0  | 0  | 0  | 2  | 65  |
| WW                       | 13                                        | 3  | 50 | 3  | 0  | 3  | 3  | 4  | 0  | 0  | 53  |
| EF                       | 0                                         | 7  | 22 | 48 | 2  | 0  | 5  | 0  | 0  | 0  | 62  |
| CB                       | 1                                         | 0  | 0  | 1  | 62 | 0  | 7  | 0  | 2  | 0  | 82  |
| CR                       | 0                                         | 0  | 10 | 3  | 0  | 37 | 0  | 3  | 0  | 3  | 72  |
| GW                       | 0                                         | 0  | 2  | 2  | 0  | 0  | 64 | 0  | 0  | 0  | 86  |
| BJ                       | 2                                         | 0  | 3  | 0  | 1  | 6  | 1  | 76 | 2  | 0  | 78  |
| CS                       | 5                                         | 0  | 5  | 0  | 3  | 0  | 0  | 7  | 54 | 0  | 79  |
| WD                       | 0                                         | 4  | 10 | 0  | 2  | 0  | 0  | 8  | 1  | 31 | 67  |
| Rel.                     | 69                                        | 75 | 46 | 68 | 78 | 79 | 80 | 74 | 87 | 86 |     |
| Overall Accuracy: 85.88% |                                           |    |    |    |    |    |    |    |    |    |     |

Analysis of GFCC(39) Features



Accuracy of GFCC(39) Features

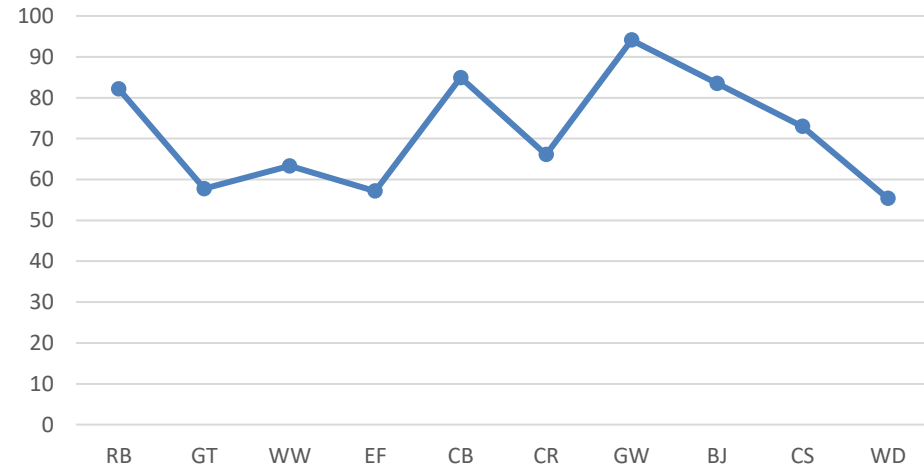
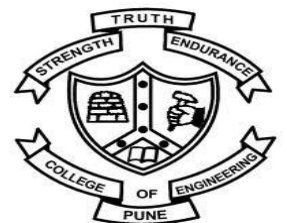


Fig 10: Analysis and Accuracy of GFCC (39) Features



Table 10: Overall Accuracy of Feature Extraction

| Sr No. | Feature vector  | MFCC  | LFCC  | GFCC  |
|--------|-----------------|-------|-------|-------|
| 1      | 13 coefficients | 82.38 | 85.53 | 87.64 |
| 2      | 26 coefficients | 81.33 | 85.49 | 87.44 |
| 3      | 39 coefficients | 82.21 | 85.88 | 85.88 |



# Feature Selection

- Feature selection selects only the relevant features and discards all irrelevant and noisy feature.
- Improves the efficiency and accuracy of the model

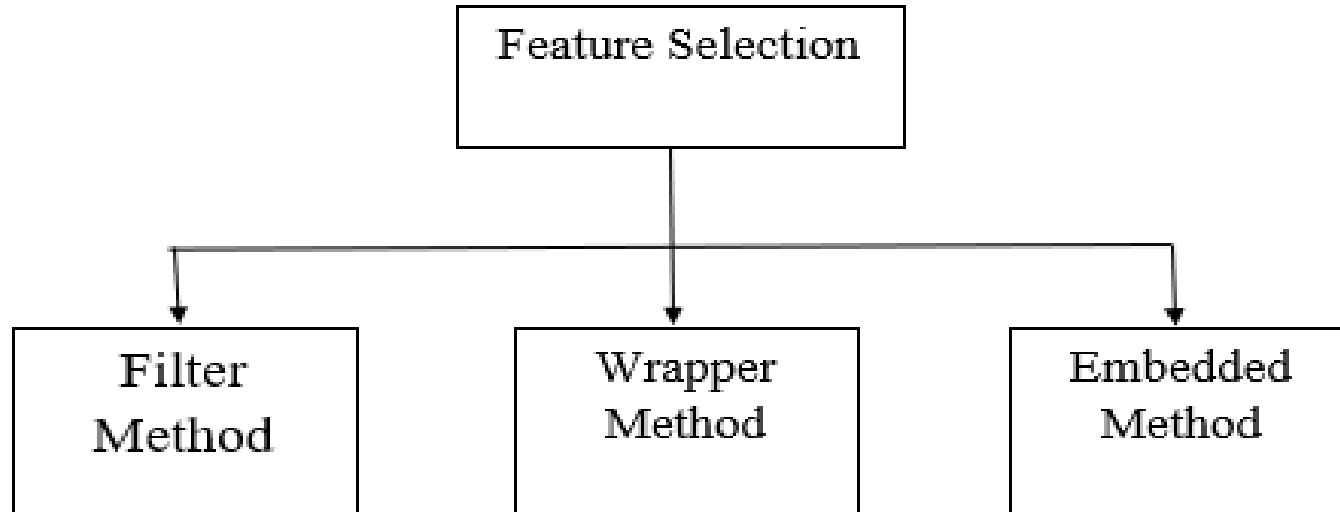


Fig 11: Block Diagram of Feature selection methods

# Recursive Feature Elimination (RFE)

- RFE generally removes the least important features and retains the remaining feature.
- The most commonly used technique under wrapper method are Forward selection, Backward elimination, Bi-directional elimination.
- Recursive Feature Elimination (RFE) algorithm select features recursively.

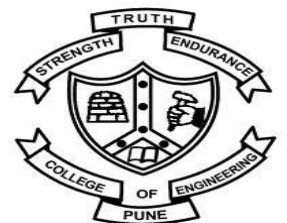


Table 11: Feature Selection

| Coefficient Number | Selected Feature | Coefficient Number | Selected Feature | Coefficient Number | Selected Feature |
|--------------------|------------------|--------------------|------------------|--------------------|------------------|
| 1                  | True             | 14                 | True             | 27                 | False            |
| 2                  | True             | 15                 | True             | 28                 | False            |
| 3                  | True             | 16                 | True             | 29                 | False            |
| 4                  | False            | 17                 | True             | 30                 | False            |
| 5                  | True             | 18                 | True             | 31                 | False            |
| 6                  | True             | 19                 | True             | 32                 | False            |
| 7                  | True             | 20                 | True             | 33                 | True             |
| 8                  | False            | 21                 | True             | 34                 | True             |
| 9                  | True             | 22                 | True             | 35                 | True             |
| 10                 | True             | 23                 | True             | 36                 | True             |
| 11                 | True             | 24                 | True             | 37                 | False            |
| 12                 | False            | 25                 | True             | 38                 | False            |
| 13                 | True             | 26                 | True             | 39                 | False            |

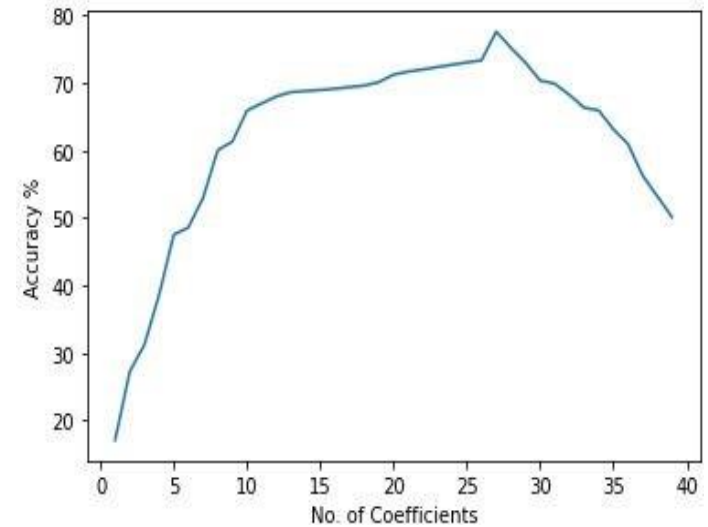
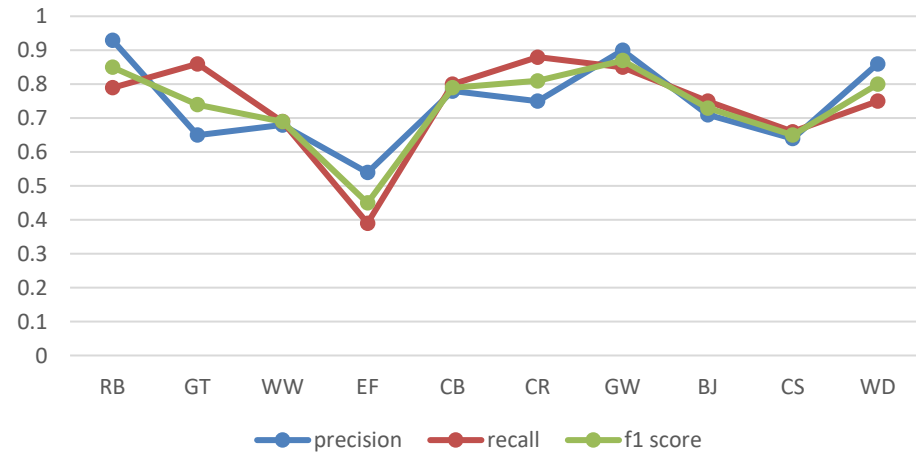


Figure 12: Accuracy vs number of features selected

Table 12: Recognition of result with MFCC (27) features

|                          | Number of correctly recognized bird sound in the class |    |    |    |    |    |    |    |    |    | RSA |
|--------------------------|--------------------------------------------------------|----|----|----|----|----|----|----|----|----|-----|
|                          | RB                                                     | GT | WW | EF | CB | CR | GW | BJ | CS | WD |     |
| RB                       | 56                                                     | 0  | 5  | 0  | 0  | 0  | 0  | 8  | 1  | 1  | 85  |
| GT                       | 0                                                      | 61 | 0  | 2  | 3  | 2  | 1  | 0  | 1  | 1  | 74  |
| WW                       | 3                                                      | 2  | 59 | 10 | 1  | 2  | 3  | 2  | 2  | 1  | 69  |
| EF                       | 0                                                      | 21 | 21 | 30 | 2  | 2  | 0  | 0  | 0  | 0  | 45  |
| CB                       | 0                                                      | 1  | 0  | 5  | 70 | 1  | 1  | 2  | 7  | 0  | 79  |
| CR                       | 0                                                      | 3  | 0  | 1  | 0  | 43 | 0  | 1  | 0  | 1  | 81  |
| GW                       | 0                                                      | 0  | 1  | 5  | 4  | 0  | 61 | 1  | 0  | 0  | 87  |
| BJ                       | 0                                                      | 1  | 1  | 0  | 1  | 5  | 2  | 59 | 10 | 0  | 73  |
| CS                       | 1                                                      | 4  | 0  | 1  | 9  | 0  | 0  | 6  | 44 | 0  | 65  |
| WD                       | 0                                                      | 1  | 0  | 2  | 0  | 2  | 0  | 4  | 4  | 38 | 80  |
| Rel.                     | 93                                                     | 65 | 68 | 54 | 78 | 75 | 90 | 71 | 64 | 86 |     |
| Overall Accuracy: 73.58% |                                                        |    |    |    |    |    |    |    |    |    |     |

Analysis of MFCC(27) Features



Accuracy of MFCC(27) Features

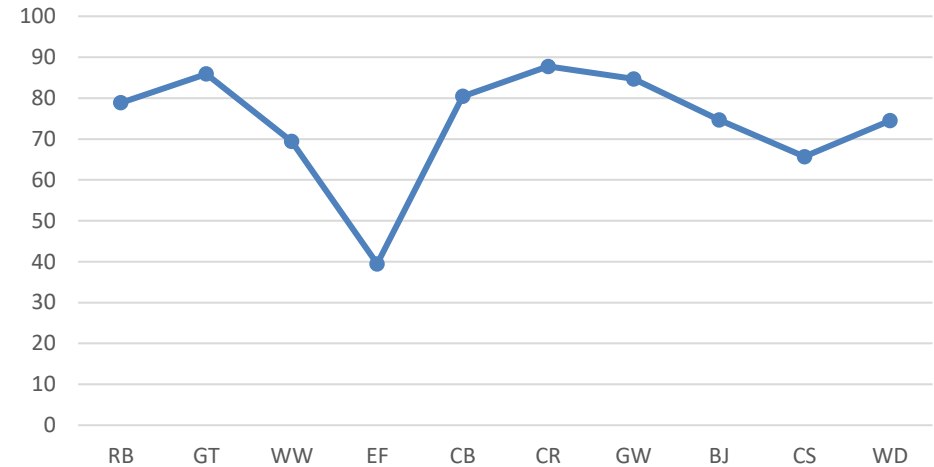
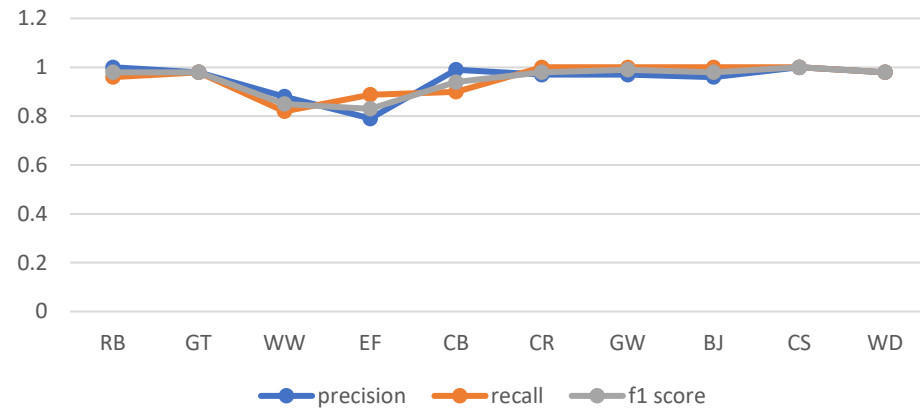


Fig 13 : Analysis and Accuracy of MFCC (27) Features

Table 13: Recognition of result with MFCC (27) features with KNN algorithm

|                          | Number of correctly recognized bird sound in the class |    |    |    |    |    |    |    |     |    | RSA |
|--------------------------|--------------------------------------------------------|----|----|----|----|----|----|----|-----|----|-----|
|                          | RB                                                     | GT | WW | EF | CB | CR | GW | BJ | CS  | WD |     |
| RB                       | 72                                                     | 1  | 0  | 0  | 0  | 0  | 0  | 2  | 0   | 1  | 96  |
| GT                       | 0                                                      | 65 | 0  | 4  | 0  | 0  | 1  | 0  | 0   | 0  | 98  |
| WW                       | 0                                                      | 0  | 61 | 12 | 0  | 1  | 0  | 0  | 0   | 0  | 82  |
| EF                       | 0                                                      | 1  | 8  | 72 | 1  | 0  | 0  | 0  | 0   | 0  | 88  |
| CB                       | 0                                                      | 0  | 0  | 7  | 77 | 1  | 1  | 0  | 0   | 0  | 90  |
| CR                       | 0                                                      | 0  | 0  | 0  | 0  | 59 | 0  | 0  | 0   | 0  | 100 |
| GW                       | 0                                                      | 0  | 0  | 0  | 0  | 0  | 69 | 1  | 0   | 0  | 100 |
| BJ                       | 0                                                      | 0  | 0  | 0  | 0  | 0  | 0  | 75 | 0   | 0  | 100 |
| CS                       | 0                                                      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 67  | 0  | 100 |
| WD                       | 0                                                      | 0  | 0  | 0  | 0  | 0  | 0  | 1  | 0   | 54 | 98  |
| Rel.                     | 100                                                    | 98 | 88 | 79 | 99 | 97 | 97 | 96 | 100 | 98 |     |
| Overall Accuracy: 94.70% |                                                        |    |    |    |    |    |    |    |     |    |     |

Analysis of MFCC(27) Features  
with KNN Algorithm



Accuracy of MFCC(27) Features  
with KNN Algorithm

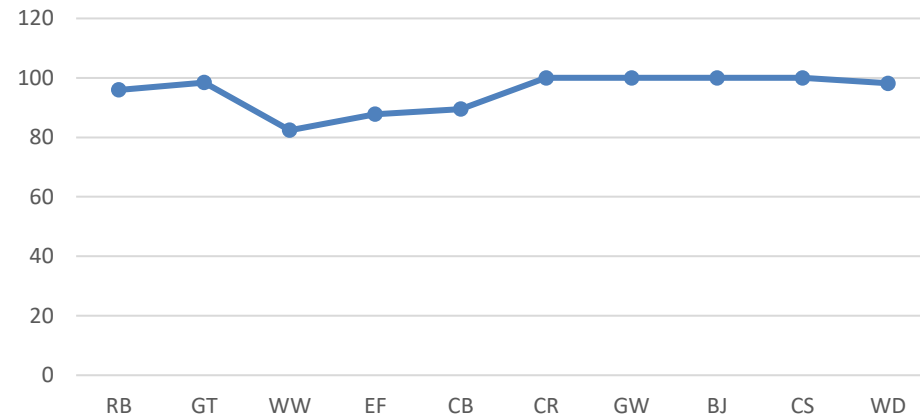


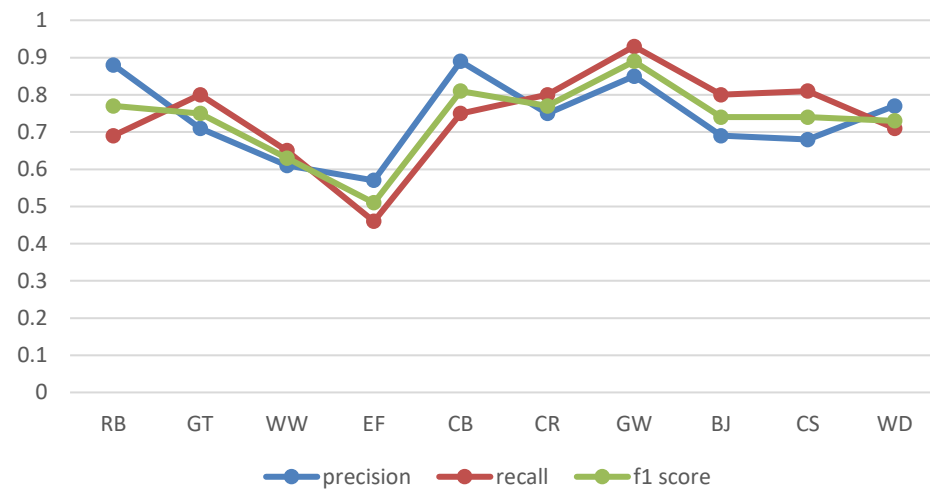
Fig 14 : Analysis and Accuracy of MFCC (27) Features with KNN Algorithm



Table 14 :Recognition of result with LFCC (27) features

|                          | Number of correctly recognized bird sound in the class |    |    |    |    |    |    |    |    |    | RSA |
|--------------------------|--------------------------------------------------------|----|----|----|----|----|----|----|----|----|-----|
|                          | RB                                                     | GT | WW | EF | CB | CR | GW | BJ | CS | WD |     |
| RB                       | 49                                                     | 0  | 5  | 0  | 1  | 5  | 1  | 7  | 3  | 0  | 69  |
| GT                       | 0                                                      | 57 | 5  | 5  | 1  | 0  | 1  | 1  | 0  | 1  | 80  |
| WW                       | 1                                                      | 3  | 55 | 12 | 0  | 2  | 5  | 2  | 4  | 1  | 65  |
| EF                       | 0                                                      | 1  | 18 | 35 | 5  | 1  | 0  | 4  | 0  | 0  | 46  |
| CB                       | 0                                                      | 3  | 0  | 7  | 65 | 1  | 0  | 0  | 0  | 0  | 75  |
| CR                       | 0                                                      | 1  | 0  | 0  | 0  | 55 | 0  | 0  | 0  | 0  | 80  |
| GW                       | 1                                                      | 1  | 2  | 0  | 0  | 0  | 63 | 2  | 0  | 0  | 93  |
| BJ                       | 0                                                      | 0  | 0  | 0  | 0  | 0  | 3  | 54 | 1  | 0  | 80  |
| CS                       | 2                                                      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 54 | 0  | 91  |
| WD                       | 2                                                      | 1  | 0  | 1  | 2  | 0  | 0  | 5  | 4  | 36 | 71  |
| Rel.                     | 88                                                     | 71 | 61 | 57 | 89 | 75 | 85 | 69 | 68 | 77 |     |
| Overall Accuracy: 87.56% |                                                        |    |    |    |    |    |    |    |    |    |     |

Analysis of LFCC(27) Features



Accuracy of LFCC(27) Features

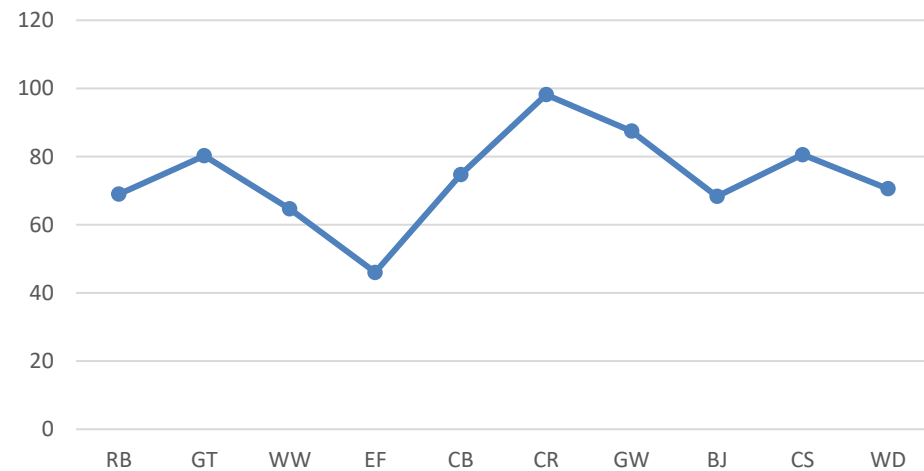
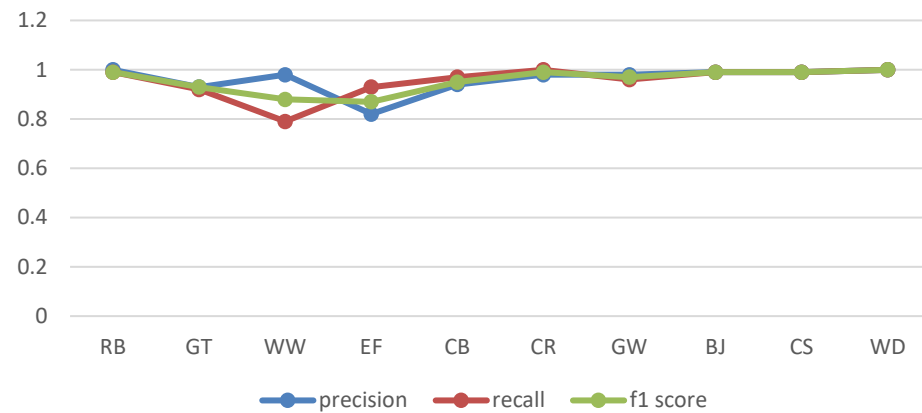


Fig 15: Analysis and Accuracy of LFCC (27) Features

Table 15: Recognition of result with LFCC (27) features with KNN algorithm

|                          | Number of correctly recognized bird sound in the class |    |    |    |    |    |    |    |    |     | RSA |
|--------------------------|--------------------------------------------------------|----|----|----|----|----|----|----|----|-----|-----|
|                          | RB                                                     | GT | WW | EF | CB | CR | GW | BJ | CS | WD  |     |
| RB                       | 66                                                     | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 99  |
| GT                       | 3                                                      | 71 | 1  | 5  | 0  | 0  | 0  | 0  | 0  | 0   | 92  |
| WW                       | 0                                                      | 0  | 58 | 13 | 0  | 1  | 0  | 0  | 0  | 0   | 79  |
| EF                       | 0                                                      | 2  | 0  | 90 | 6  | 4  | 0  | 4  | 0  | 0   | 93  |
| CB                       | 0                                                      | 0  | 0  | 5  | 72 | 0  | 0  | 0  | 0  | 0   | 97  |
| CR                       | 0                                                      | 0  | 0  | 0  | 0  | 54 | 0  | 0  | 0  | 0   | 100 |
| GW                       | 0                                                      | 2  | 0  | 0  | 0  | 0  | 54 | 0  | 0  | 0   | 96  |
| BJ                       | 0                                                      | 0  | 0  | 0  | 0  | 0  | 0  | 78 | 1  | 0   | 99  |
| CS                       | 0                                                      | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 67 | 0   | 99  |
| WD                       | 0                                                      | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 63  | 100 |
| Rel.                     | 95                                                     | 93 | 98 | 82 | 94 | 98 | 98 | 99 | 99 | 100 |     |
| Overall Accuracy: 95.05% |                                                        |    |    |    |    |    |    |    |    |     |     |

Analysis of LFCC(27) Features  
with KNN Algorithm



Accuracy of LFCC(27) Features  
with KNN Algorithm

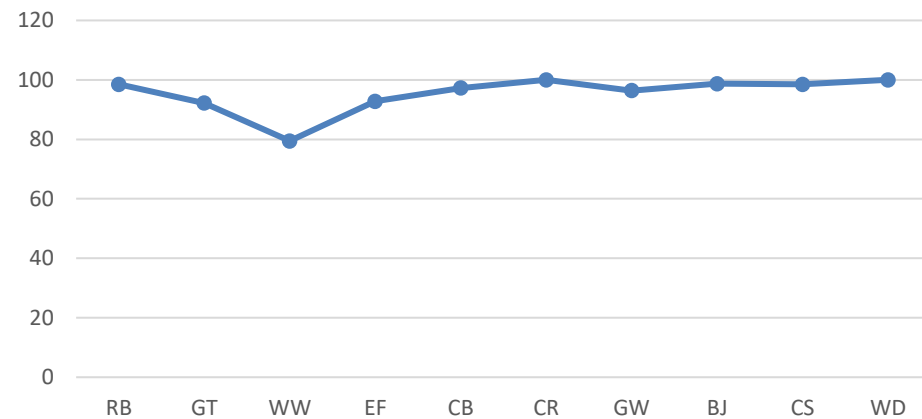
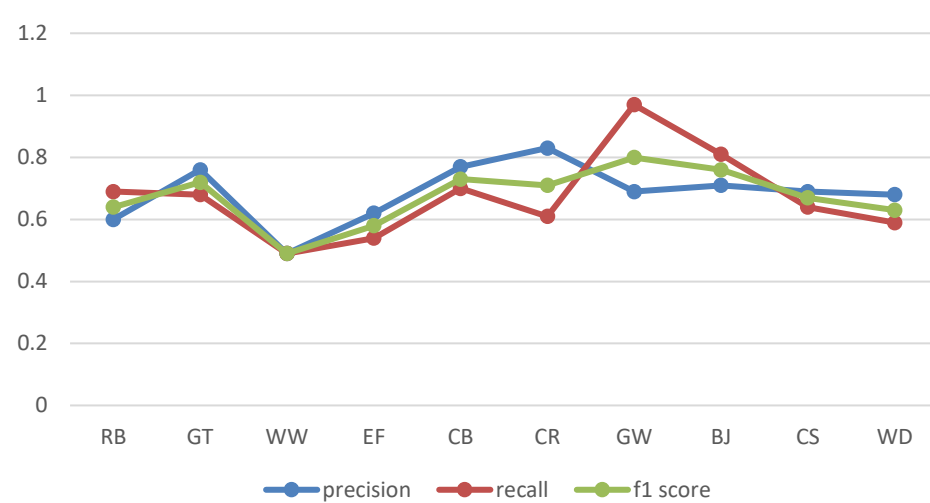


Fig 16 : Analysis and Accuracy of LFCC (27) Features with KNN Algorithm

Table 16 :Recognition of result with GFCC (27) features

|                          | Number of correctly recognized bird sound in the class |    |    |    |    |    |    |    |    |    | RSA |
|--------------------------|--------------------------------------------------------|----|----|----|----|----|----|----|----|----|-----|
|                          | RB                                                     | GT | WW | EF | CB | CR | GW | BJ | CS | WD |     |
| RB                       | 49                                                     | 0  | 6  | 2  | 0  | 1  | 0  | 7  | 7  | 0  | 69  |
| GT                       | 0                                                      | 48 | 3  | 7  | 7  | 0  | 0  | 0  | 0  | 6  | 68  |
| WW                       | 19                                                     | 4  | 42 | 5  | 1  | 3  | 6  | 5  | 0  | 0  | 49  |
| EF                       | 0                                                      | 5  | 24 | 41 | 2  | 0  | 0  | 0  | 0  | 0  | 54  |
| CB                       | 1                                                      | 1  | 0  | 3  | 82 | 1  | 2  | 0  | 0  | 0  | 70  |
| CR                       | 1                                                      | 0  | 5  | 3  | 0  | 30 | 0  | 3  | 0  | 7  | 61  |
| GW                       | 0                                                      | 0  | 0  | 0  | 0  | 0  | 60 | 1  | 0  | 0  | 97  |
| BJ                       | 3                                                      | 0  | 0  | 0  | 2  | 2  | 2  | 64 | 6  | 1  | 81  |
| CS                       | 8                                                      | 0  | 2  | 3  | 4  | 0  | 0  | 7  | 43 | 0  | 64  |
| WD                       | 0                                                      | 5  | 3  | 1  | 3  | 1  | 4  | 4  | 0  | 30 | 59  |
| Rel.                     | 60                                                     | 76 | 49 | 62 | 77 | 83 | 69 | 71 | 69 | 68 |     |
| Overall Accuracy: 85.22% |                                                        |    |    |    |    |    |    |    |    |    |     |

Analysis of GFCC(27) Features



Accuracy of GFCC(27) Features

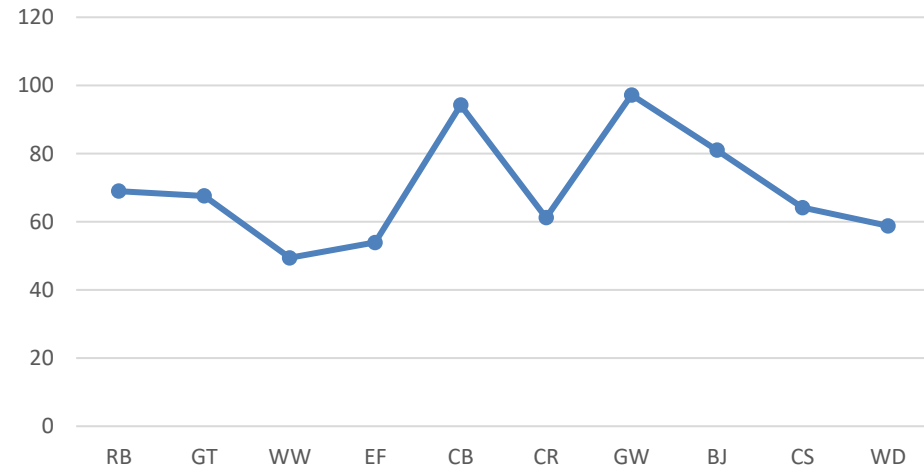
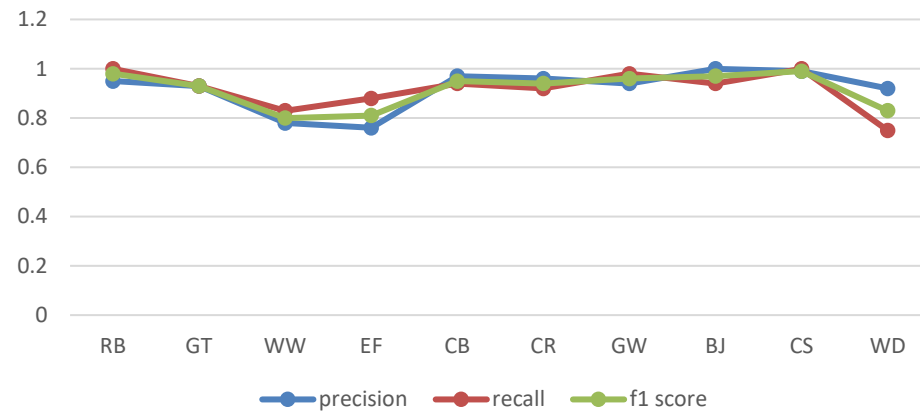


Fig 17 : Analysis and Accuracy of GFCC (27) Features

Table 17: Recognition of result with GFCC (27) features with KNN algorithm

|                          | Number of correctly recognized bird sound in the class |    |    |    |    |    |    |     |    |    | RSA |
|--------------------------|--------------------------------------------------------|----|----|----|----|----|----|-----|----|----|-----|
|                          | RB                                                     | GT | WW | EF | CB | CR | GW | BJ  | CS | WD |     |
| RB                       | 62                                                     | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 0  | 100 |
| GT                       | 0                                                      | 69 | 2  | 3  | 0  | 0  | 0  | 0   | 0  | 0  | 93  |
| WW                       | 0                                                      | 2  | 57 | 10 | 0  | 1  | 0  | 0   | 0  | 0  | 83  |
| EF                       | 0                                                      | 0  | 6  | 65 | 3  | 0  | 0  | 0   | 0  | 0  | 88  |
| CB                       | 1                                                      | 0  | 0  | 5  | 87 | 0  | 2  | 0   | 0  | 0  | 94  |
| CR                       | 0                                                      | 0  | 0  | 0  | 0  | 55 | 1  | 0   | 0  | 1  | 92  |
| GW                       | 0                                                      | 0  | 0  | 0  | 0  | 0  | 61 | 0   | 0  | 0  | 98  |
| BJ                       | 0                                                      | 0  | 0  | 0  | 0  | 0  | 3  | 73  | 1  | 0  | 94  |
| CS                       | 0                                                      | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 71 | 0  | 100 |
| WD                       | 1                                                      | 3  | 7  | 3  | 0  | 2  | 0  | 0   | 0  | 49 | 75  |
| Rel.                     | 95                                                     | 93 | 78 | 76 | 97 | 96 | 94 | 100 | 99 | 92 |     |
| Overall Accuracy: 91.66% |                                                        |    |    |    |    |    |    |     |    |    |     |

Analysys of GFCC(27) Features  
with KNN Algorithm



Accuracy of GFCC (27) Features  
with KNN Algorithm

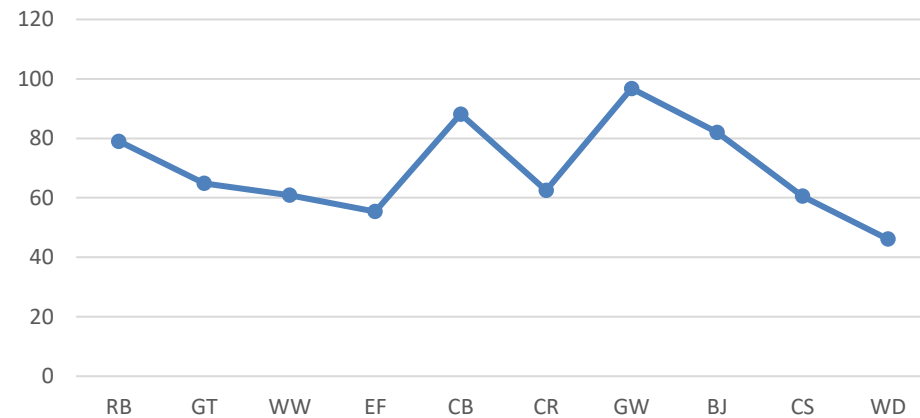
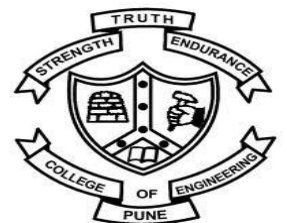


Fig 18 : Analysis and Accuracy of GFCC (27) Features with KNN Algorithm



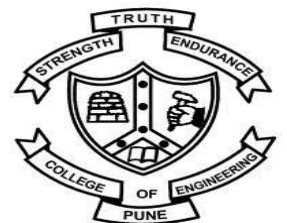
Table 18 :Overall Accuracy of Feature Selection

| Sr. No. | Feature set | KNN   | SVM   |
|---------|-------------|-------|-------|
| 1       | MFCC        | 94.7  | 73.58 |
| 2       | LFCC        | 95.05 | 87.56 |
| 3       | GFCC        | 91.66 | 85.22 |



# K-Nearest Neighbors

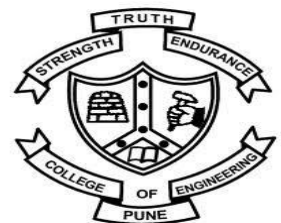
- Supervised machine learning algorithm used for regression and classification.
- Each query point considers the closest k value and predicts the maximum possible class.
- It is also used for identifying similar objects.
- For each bird audio signal, the mean value of the feature is extracted to form a vector.



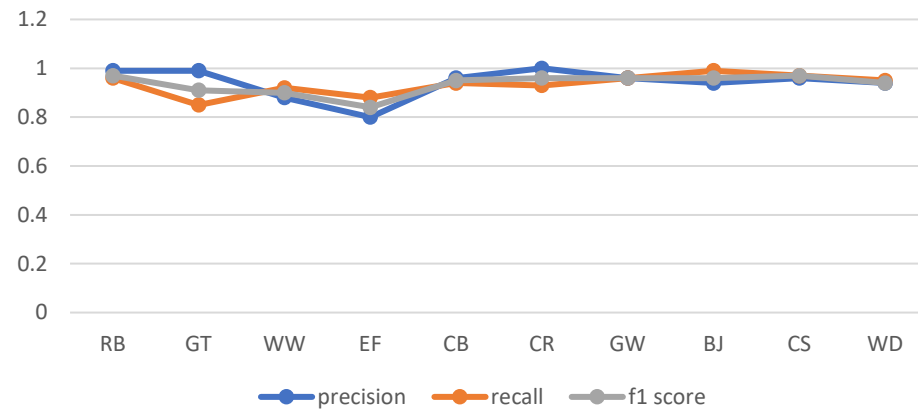
# Result of MFCC

Table 19: Recognition of result with MFCC (13) features with KNN algorithm

|                          | Number of correctly recognized bird sound |    |    |    |    |     |    |    |    |    | RSA |
|--------------------------|-------------------------------------------|----|----|----|----|-----|----|----|----|----|-----|
|                          | RB                                        | GT | WW | EF | CB | CR  | GW | BJ | CS | WD |     |
| RB                       | 69                                        | 0  | 0  | 0  | 1  | 0   | 0  | 0  | 2  | 0  | 97  |
| GT                       | 0                                         | 66 | 1  | 8  | 0  | 0   | 0  | 0  | 0  | 3  | 91  |
| WW                       | 0                                         | 1  | 72 | 5  | 0  | 0   | 0  | 0  | 0  | 0  | 90  |
| EF                       | 0                                         | 0  | 9  | 68 | 0  | 0   | 0  | 0  | 0  | 0  | 84  |
| CB                       | 1                                         | 0  | 0  | 2  | 68 | 0   | 1  | 0  | 0  | 0  | 95  |
| CR                       | 0                                         | 0  | 0  | 2  | 0  | 54  | 1  | 0  | 0  | 1  | 96  |
| GW                       | 0                                         | 0  | 0  | 0  | 0  | 0   | 54 | 2  | 0  | 0  | 96  |
| BJ                       | 0                                         | 0  | 0  | 0  | 0  | 0   | 0  | 79 | 1  | 0  | 96  |
| CS                       | 0                                         | 0  | 0  | 0  | 2  | 0   | 0  | 0  | 74 | 0  | 97  |
| WD                       | 0                                         | 0  | 0  | 0  | 0  | 0   | 0  | 0  | 0  | 58 | 94  |
| Rel.                     | 99                                        | 99 | 88 | 80 | 96 | 100 | 96 | 94 | 96 | 94 |     |
| Overall Accuracy: 91.13% |                                           |    |    |    |    |     |    |    |    |    |     |



Analysis of MFCC(13) Features  
with KNN Algorithm



Accuracy of MFCC(13) Features  
with KNN Algorithm

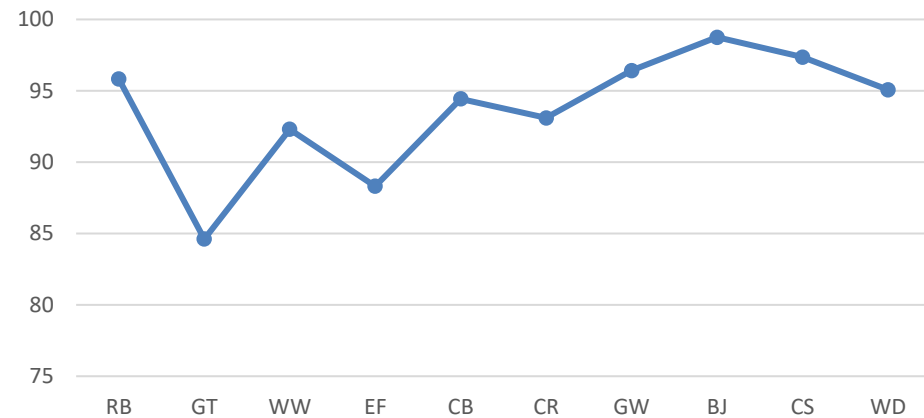
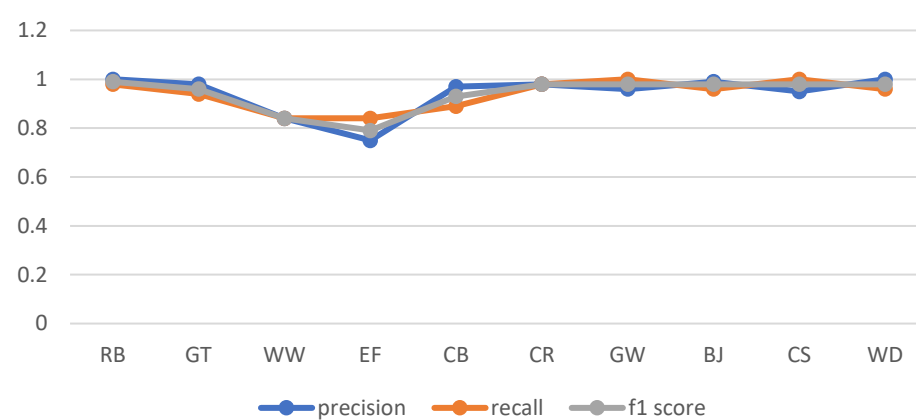


Fig 19 : Analysis and Accuracy of MFCC (13) Features with KNN Algorithm

Table 20: Recognition of result with MFCC (26) features with KNN algorithm

|                          | Number of correctly recognized bird sound |    |    |    |    |    |    |    |    |    | RSA |
|--------------------------|-------------------------------------------|----|----|----|----|----|----|----|----|----|-----|
|                          | RB                                        | GT | WW | EF | CB | CR | GW | BJ | CS | WD |     |
| RB                       | 62                                        | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 99  |
| GT                       | 0                                         | 81 | 1  | 4  | 0  | 0  | 0  | 0  | 0  | 0  | 96  |
| WW                       | 0                                         | 1  | 61 | 11 | 0  | 0  | 0  | 0  | 0  | 0  | 84  |
| EF                       | 0                                         | 0  | 11 | 66 | 2  | 0  | 0  | 0  | 0  | 0  | 79  |
| CB                       | 0                                         | 0  | 0  | 6  | 71 | 1  | 1  | 0  | 1  | 0  | 93  |
| CR                       | 0                                         | 0  | 0  | 0  | 0  | 55 | 1  | 0  | 0  | 0  | 98  |
| GW                       | 0                                         | 0  | 0  | 0  | 0  | 0  | 69 | 0  | 0  | 0  | 98  |
| BJ                       | 0                                         | 0  | 0  | 0  | 0  | 0  | 1  | 80 | 2  | 0  | 98  |
| CS                       | 0                                         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 63 | 0  | 98  |
| WD                       | 0                                         | 0  | 0  | 0  | 1  | 0  | 0  | 1  | 0  | 54 | 98  |
| Rel.                     | 100                                       | 98 | 84 | 75 | 97 | 98 | 96 | 99 | 95 | 1  |     |
| Overall Accuracy: 91.50% |                                           |    |    |    |    |    |    |    |    |    |     |

Analysis of MFCC(26) Features  
With KNN Algorithm



Accuracy of MFCC(26) Features  
with KNN Algorithm

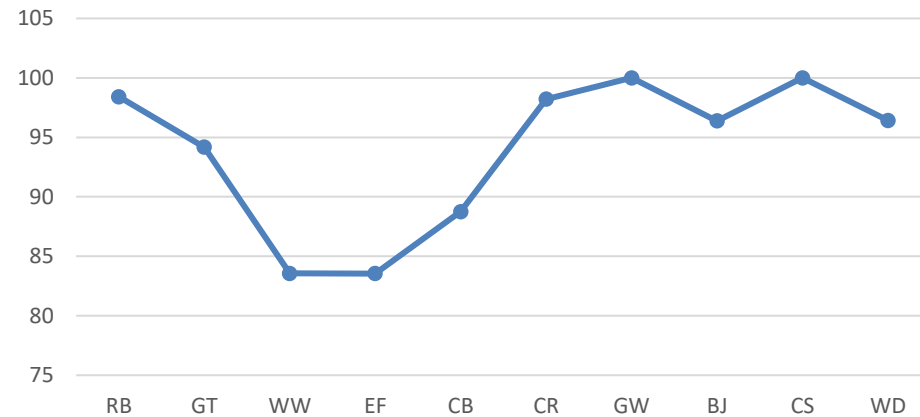
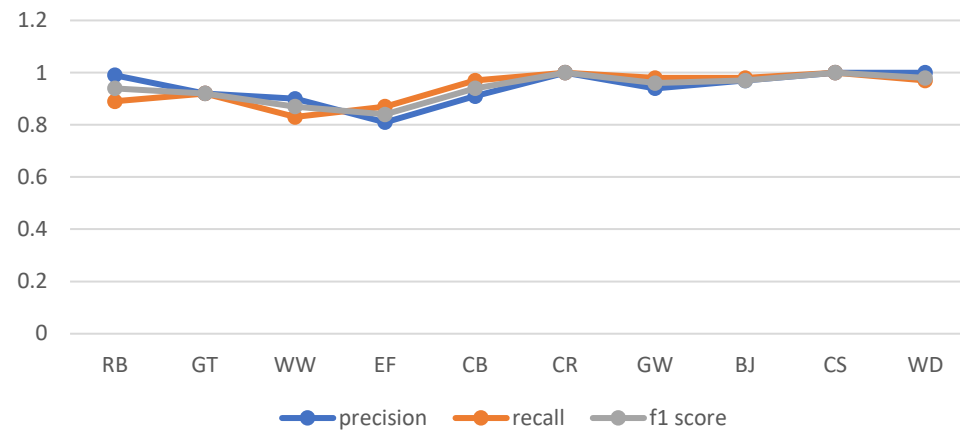


Fig 20 : Analysis and Accuracy of MFCC (26) Features with KNN Algorithm

Table 21: Recognition of result with MFCC (39) features with KNN algorithm

|                          | Number of correctly recognized bird sound |    |    |    |    |     |    |    |     |     | RSA |
|--------------------------|-------------------------------------------|----|----|----|----|-----|----|----|-----|-----|-----|
|                          | RB                                        | GT | WW | EF | CB | CR  | GW | BJ | CS  | WD  |     |
| RB                       | 68                                        | 2  | 0  | 0  | 5  | 0   | 0  | 1  | 0   | 0   | 94  |
| GT                       | 0                                         | 57 | 0  | 5  | 0  | 0   | 0  | 0  | 0   | 0   | 92  |
| WW                       | 0                                         | 1  | 65 | 12 | 0  | 0   | 0  | 0  | 0   | 0   | 87  |
| EF                       | 0                                         | 1  | 6  | 77 | 2  | 0   | 3  | 0  | 0   | 0   | 84  |
| CB                       | 0                                         | 0  | 0  | 1  | 72 | 0   | 1  | 0  | 0   | 0   | 94  |
| CR                       | 0                                         | 0  | 0  | 0  | 0  | 48  | 0  | 0  | 0   | 0   | 100 |
| GW                       | 0                                         | 0  | 0  | 0  | 0  | 0   | 60 | 1  | 0   | 0   | 96  |
| BJ                       | 0                                         | 1  | 1  | 0  | 0  | 0   | 0  | 85 | 0   | 0   | 97  |
| CS                       | 0                                         | 0  | 0  | 0  | 0  | 0   | 0  | 0  | 74  | 0   | 100 |
| WD                       | 1                                         | 0  | 0  | 0  | 0  | 0   | 0  | 1  | 0   | 57  | 98  |
| Rel.                     | 99                                        | 92 | 90 | 81 | 91 | 100 | 94 | 97 | 100 | 100 |     |
| Overall Accuracy: 93.64% |                                           |    |    |    |    |     |    |    |     |     |     |

Analysis of MFCC(39) Features  
With KNN Algorithm



Accuracy of MFCC(39) Features  
with KNN Algorithm

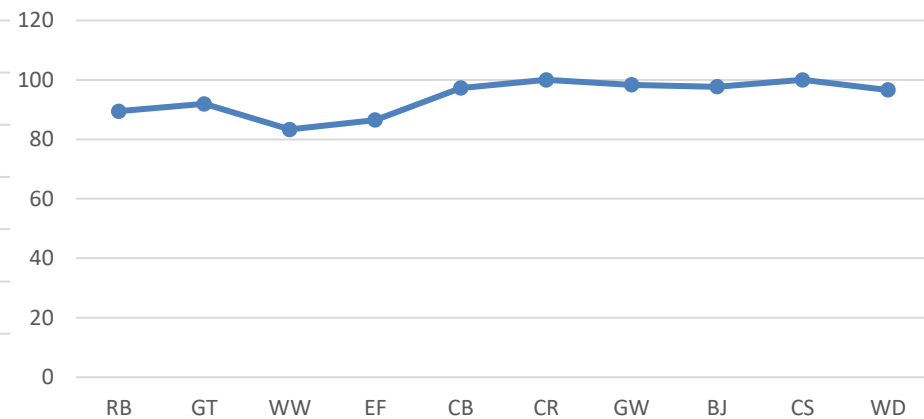


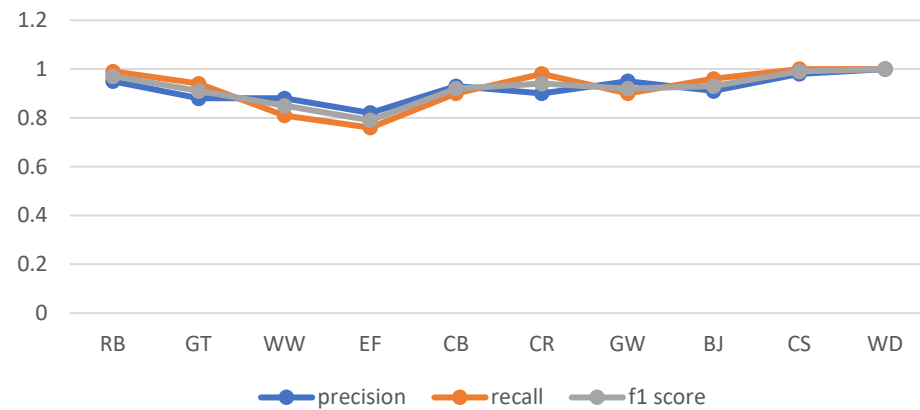
Fig 21 : Analysis and Accuracy of MFCC (39) Features with KNN Algorithm



Table 22: Recognition of result with LFCC (13) features with KNN algorithm

|                          | Number of correctly recognized bird sound |    |    |    |    |    |    |    |    |     | RSA |
|--------------------------|-------------------------------------------|----|----|----|----|----|----|----|----|-----|-----|
|                          | RB                                        | GT | WW | EF | CB | CR | GW | BJ | CS | WD  |     |
| RB                       | 69                                        | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 97  |
| GT                       | 3                                         | 65 | 1  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 91  |
| WW                       | 0                                         | 3  | 61 | 8  | 0  | 1  | 0  | 2  | 0  | 0   | 85  |
| EF                       | 0                                         | 0  | 5  | 61 | 6  | 4  | 0  | 4  | 0  | 0   | 79  |
| CB                       | 0                                         | 3  | 0  | 5  | 83 | 1  | 0  | 0  | 0  | 0   | 92  |
| CR                       | 0                                         | 1  | 0  | 0  | 0  | 55 | 0  | 0  | 0  | 0   | 94  |
| GW                       | 1                                         | 1  | 2  | 0  | 0  | 0  | 55 | 2  | 0  | 0   | 92  |
| BJ                       | 0                                         | 0  | 0  | 0  | 0  | 0  | 3  | 86 | 1  | 0   | 93  |
| CS                       | 0                                         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 63 | 0   | 99  |
| WD                       | 0                                         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 52  | 100 |
| Rel.                     | 95                                        | 88 | 88 | 82 | 93 | 90 | 95 | 91 | 98 | 100 |     |
| Overall Accuracy: 91.80% |                                           |    |    |    |    |    |    |    |    |     |     |

Analysis of LFCC(13) Features  
with KNN Algorithm



Accuracy of LFCC(13) Features  
with KNN Algorithm

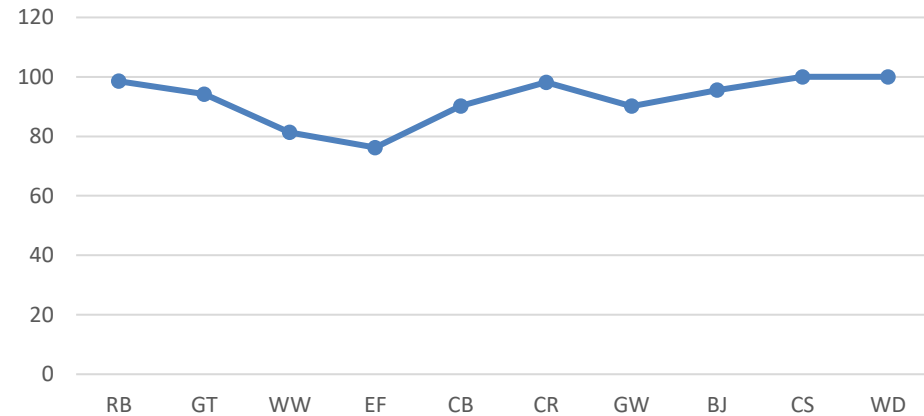
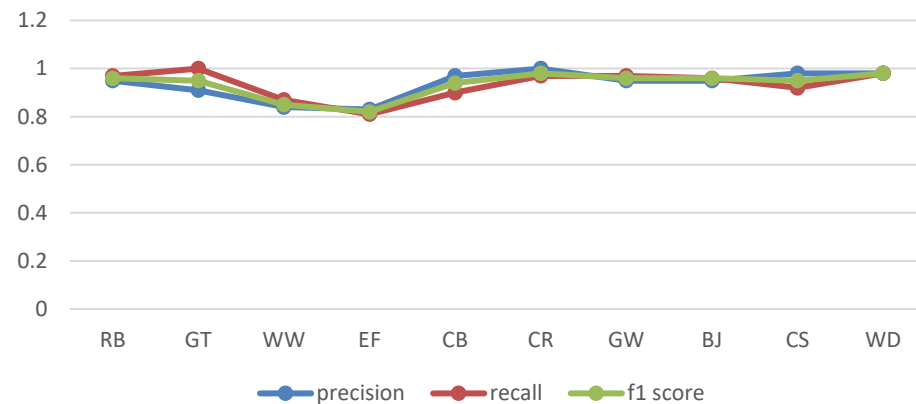


Fig 22 : Analysis and Accuracy of LFCC (13) Features with KNN Algorithm

Table 23: Recognition of result with LFCC (26) features with KNN algorithm

|                          | Number of correctly recognized bird sound |    |    |    |    |     |    |    |    |    | RSA |
|--------------------------|-------------------------------------------|----|----|----|----|-----|----|----|----|----|-----|
|                          | RB                                        | GT | WW | EF | CB | CR  | GW | BJ | CS | WD |     |
| RB                       | 69                                        | 2  | 0  | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 96  |
| GT                       | 0                                         | 67 | 0  | 0  | 0  | 0   | 0  | 0  | 0  | 0  | 95  |
| WW                       | 0                                         | 1  | 67 | 7  | 0  | 0   | 0  | 2  | 0  | 0  | 85  |
| EF                       | 0                                         | 1  | 13 | 70 | 2  | 0   | 0  | 0  | 0  | 0  | 82  |
| CB                       | 1                                         | 0  | 0  | 6  | 73 | 0   | 0  | 0  | 1  | 0  | 94  |
| CR                       | 0                                         | 2  | 0  | 0  | 0  | 61  | 0  | 0  | 0  | 0  | 98  |
| GW                       | 0                                         | 0  | 0  | 0  | 0  | 0   | 60 | 2  | 0  | 0  | 96  |
| BJ                       | 0                                         | 0  | 0  | 0  | 0  | 0   | 3  | 78 | 0  | 0  | 96  |
| CS                       | 3                                         | 1  | 0  | 0  | 0  | 0   | 0  | 0  | 57 | 1  | 95  |
| WD                       | 0                                         | 0  | 0  | 1  | 0  | 0   | 0  | 0  | 0  | 57 | 98  |
| Rel.                     | 95                                        | 91 | 84 | 83 | 97 | 100 | 95 | 95 | 98 | 98 |     |
| Overall Accuracy: 91.76% |                                           |    |    |    |    |     |    |    |    |    |     |

Analysis of LFCC(26) Features  
with KNN Algorithm



Accuracy of LFCC(26) Features  
with KNN Algorithm

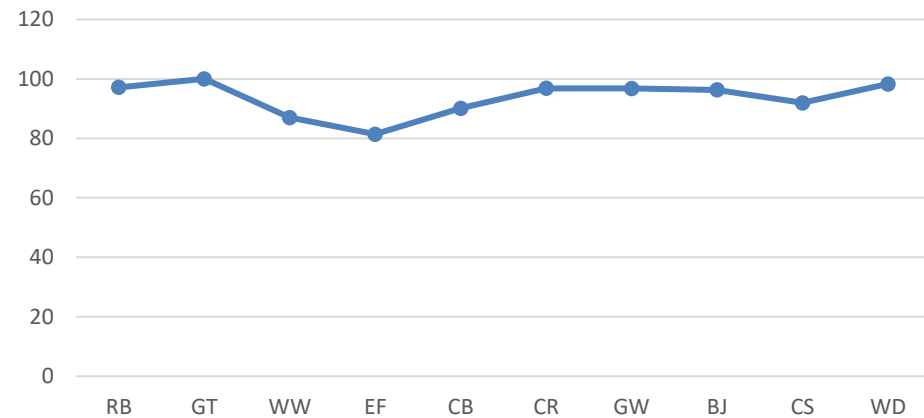
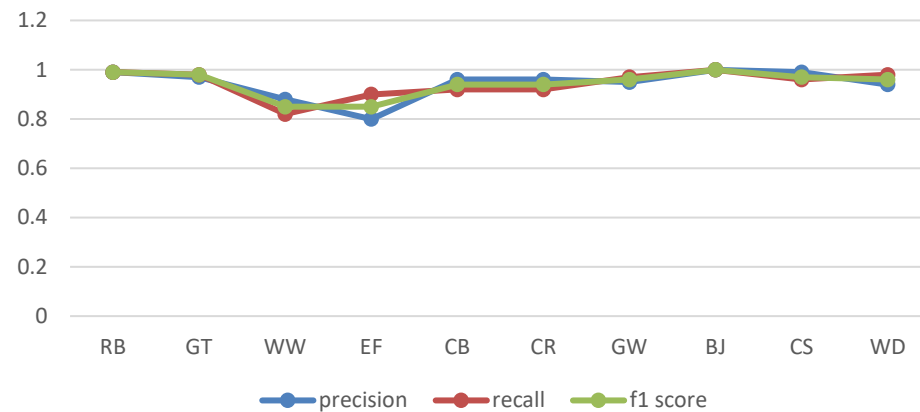


Fig 23 : Analysis and Accuracy of LFCC (26) Features with KNN Algorithm

Table 24: Recognition of result with LFCC (39) features with KNN algorithm

|                          | Number of correctly recognized bird sound |    |    |    |    |    |    |     |    |    | RSA |
|--------------------------|-------------------------------------------|----|----|----|----|----|----|-----|----|----|-----|
|                          | RB                                        | GT | WW | EF | CB | CR | GW | BJ  | CS | WD |     |
| RB                       | 77                                        | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 1  | 0  | 99  |
| GT                       | 0                                         | 63 | 1  | 0  | 0  | 0  | 0  | 0   | 0  | 0  | 98  |
| WW                       | 0                                         | 1  | 66 | 12 | 0  | 0  | 0  | 0   | 0  | 1  | 85  |
| EF                       | 0                                         | 1  | 4  | 70 | 3  | 0  | 0  | 0   | 0  | 0  | 85  |
| CB                       | 0                                         | 0  | 0  | 2  | 81 | 2  | 3  | 0   | 0  | 0  | 94  |
| CR                       | 0                                         | 0  | 2  | 2  | 0  | 49 | 0  | 0   | 0  | 0  | 94  |
| GW                       | 0                                         | 0  | 2  | 0  | 0  | 0  | 63 | 0   | 0  | 0  | 96  |
| BJ                       | 0                                         | 0  | 0  | 0  | 0  | 0  | 0  | 78  | 0  | 0  | 100 |
| CS                       | 1                                         | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 69 | 2  | 97  |
| WD                       | 0                                         | 0  | 0  | 1  | 0  | 0  | 0  | 0   | 0  | 51 | 96  |
| Rel.                     | 99                                        | 97 | 88 | 80 | 96 | 96 | 95 | 100 | 99 | 94 |     |
| Overall Accuracy: 91.90% |                                           |    |    |    |    |    |    |     |    |    |     |

Analysis of LFCC(39) Features  
with KNN Algorithm



Accuracy of LFCC(39) Features  
with KNN Algorithm

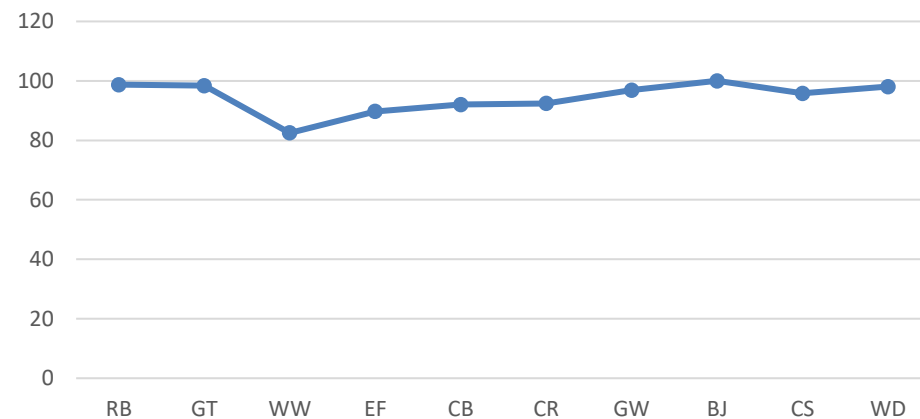
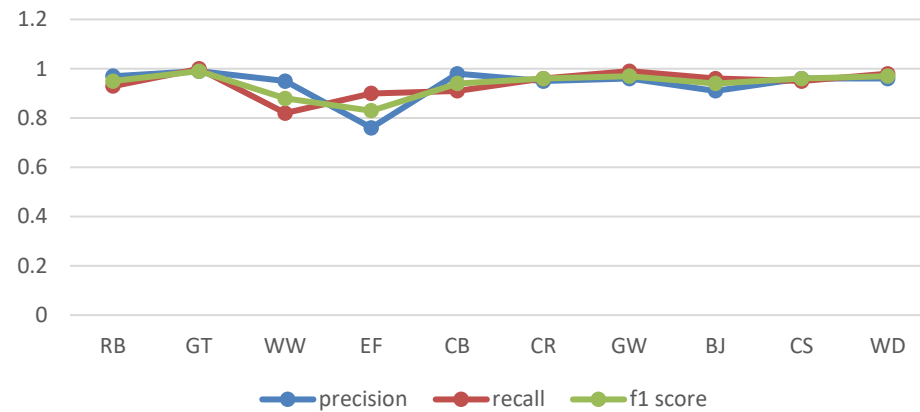


Fig 24 : Analysis and Accuracy of LFCC (39) Features with KNN Algorithm

Table 25: Recognition of result with GFCC (13) features with KNN algorithm

|                          | Number of correctly recognized bird sound |    |    |    |    |    |    |    |    |    | RSA |
|--------------------------|-------------------------------------------|----|----|----|----|----|----|----|----|----|-----|
|                          | RB                                        | GT | WW | EF | CB | CR | GW | BJ | CS | WD |     |
| RB                       | 65                                        | 0  | 0  | 0  | 0  | 1  | 0  | 2  | 2  | 0  | 95  |
| GT                       | 0                                         | 71 | 0  | 0  | 0  | 0  | 0  | 1  | 0  | 0  | 99  |
| WW                       | 0                                         | 1  | 70 | 13 | 0  | 1  | 0  | 0  | 0  | 0  | 88  |
| EF                       | 0                                         | 0  | 4  | 55 | 2  | 0  | 0  | 0  | 0  | 0  | 83  |
| CB                       | 1                                         | 0  | 0  | 4  | 82 | 1  | 2  | 0  | 0  | 0  | 94  |
| CR                       | 0                                         | 0  | 0  | 0  | 0  | 54 | 0  | 1  | 0  | 1  | 96  |
| GW                       | 0                                         | 0  | 0  | 0  | 0  | 0  | 66 | 1  | 0  | 0  | 97  |
| BJ                       | 0                                         | 0  | 0  | 0  | 0  | 0  | 1  | 80 | 1  | 1  | 94  |
| CS                       | 0                                         | 0  | 0  | 0  | 0  | 0  | 0  | 4  | 75 | 0  | 96  |
| WD                       | 1                                         | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 44 | 97  |
| Rel.                     | 97                                        | 99 | 95 | 76 | 98 | 95 | 96 | 91 | 96 | 96 |     |
| Overall Accuracy: 91.64% |                                           |    |    |    |    |    |    |    |    |    |     |

Analysis of GFCC (13) Features  
with KNN Algorithm



Accuracy of GFCC (13) Features  
with KNN Algorithm

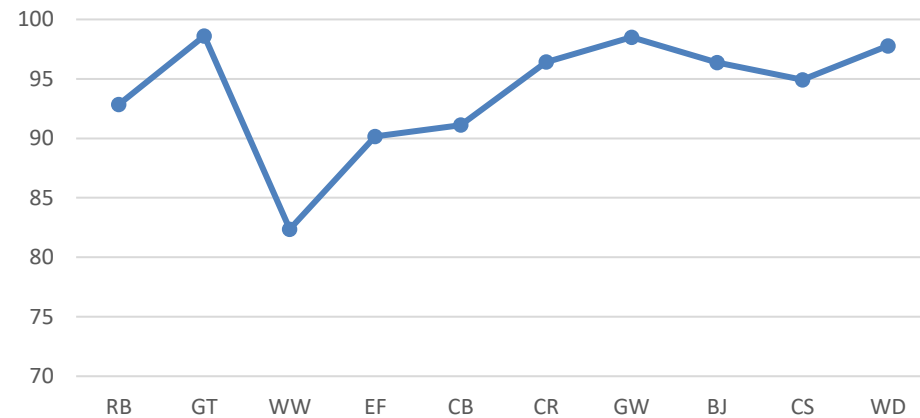


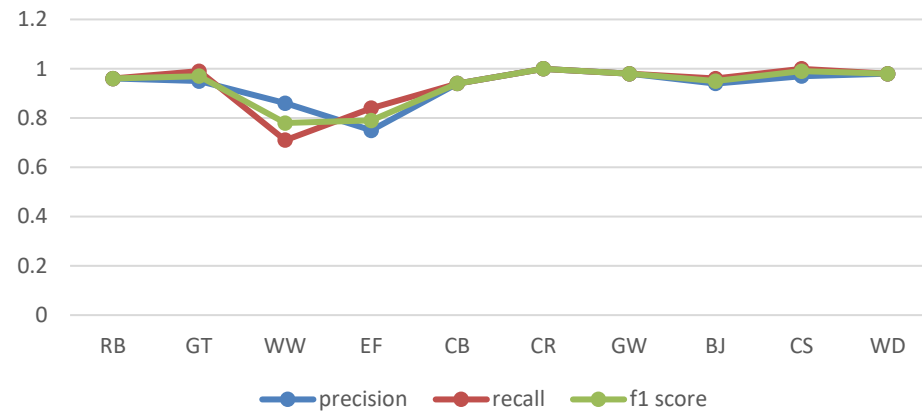
Fig 25: Analysis and Accuracy of GFCC (13) Features with KNN Algorithm



Table 26: Recognition of result with GFCC (26) features with KNN algorithm

|                          | Number of correctly recognized bird sound |    |    |    |    |    |    |     |    |    | RSA |
|--------------------------|-------------------------------------------|----|----|----|----|----|----|-----|----|----|-----|
|                          | RB                                        | GT | WW | EF | CB | CR | GW | BJ  | CS | WD |     |
| RB                       | 77                                        | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 1  | 0  | 99  |
| GT                       | 0                                         | 63 | 1  | 0  | 0  | 0  | 0  | 0   | 0  | 0  | 98  |
| WW                       | 0                                         | 1  | 66 | 12 | 0  | 0  | 0  | 0   | 0  | 1  | 85  |
| EF                       | 0                                         | 1  | 4  | 70 | 3  | 0  | 0  | 0   | 0  | 0  | 85  |
| CB                       | 0                                         | 0  | 0  | 2  | 71 | 2  | 3  | 0   | 0  | 0  | 94  |
| CR                       | 0                                         | 0  | 2  | 2  | 0  | 69 | 0  | 0   | 0  | 0  | 94  |
| GW                       | 0                                         | 0  | 3  | 0  | 0  | 0  | 63 | 0   | 0  | 0  | 96  |
| BJ                       | 0                                         | 0  | 0  | 0  | 0  | 0  | 0  | 78  | 0  | 0  | 100 |
| CS                       | 1                                         | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 69 | 2  | 97  |
| WD                       | 0                                         | 0  | 0  | 1  | 0  | 0  | 0  | 0   | 0  | 51 | 96  |
| Rel.                     | 99                                        | 97 | 88 | 80 | 96 | 96 | 95 | 100 | 99 | 94 |     |
| Overall Accuracy: 91.90% |                                           |    |    |    |    |    |    |     |    |    |     |

Analysis of GFCC(26) Features  
with KNN Algorithm



Accuracy of GFCC (26) Features  
with KNN Algorithm

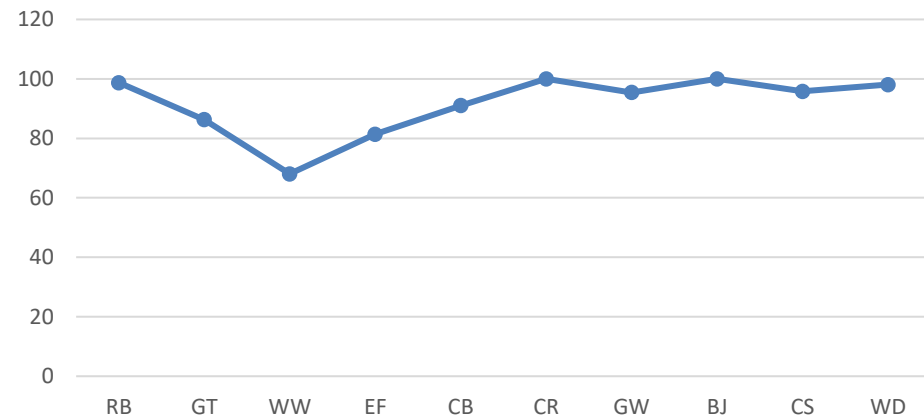
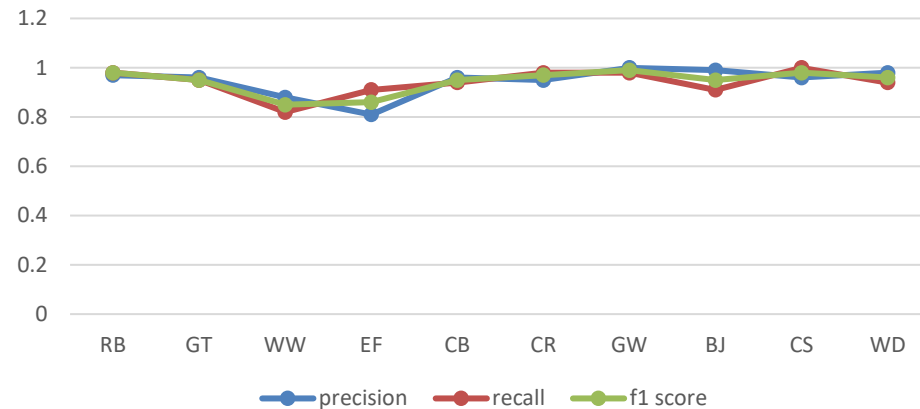


Fig 26 : Analysis and Accuracy of GFCC (26) Features with KNN Algorithm

Table 27: Recognition of result with GFCC (39) features with KNN algorithm

|                          | Number of correctly recognized bird sound |    |    |    |    |    |     |    |    |    | RSA |
|--------------------------|-------------------------------------------|----|----|----|----|----|-----|----|----|----|-----|
|                          | RB                                        | GT | WW | EF | CB | CR | GW  | BJ | CS | WD |     |
| RB                       | 64                                        | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 1  | 0  | 98  |
| GT                       | 0                                         | 71 | 0  | 2  | 0  | 2  | 0   | 0  | 0  | 0  | 95  |
| WW                       | 0                                         | 2  | 63 | 12 | 0  | 0  | 0   | 0  | 0  | 0  | 85  |
| EF                       | 0                                         | 1  | 4  | 81 | 3  | 0  | 0   | 0  | 0  | 0  | 86  |
| CB                       | 1                                         | 0  | 0  | 3  | 77 | 1  | 0   | 0  | 0  | 0  | 95  |
| CR                       | 0                                         | 0  | 0  | 0  | 0  | 60 | 0   | 0  | 0  | 1  | 97  |
| GW                       | 0                                         | 0  | 0  | 0  | 0  | 0  | 56  | 1  | 0  | 0  | 99  |
| BJ                       | 0                                         | 0  | 5  | 0  | 0  | 0  | 0   | 71 | 2  | 0  | 95  |
| CS                       | 0                                         | 0  | 0  | 0  | 0  | 0  | 0   | 0  | 75 | 0  | 98  |
| WD                       | 1                                         | 0  | 0  | 2  | 0  | 0  | 0   | 0  | 0  | 46 | 96  |
| Rel.                     | 97                                        | 96 | 88 | 81 | 96 | 95 | 100 | 99 | 96 | 98 |     |
| Overall Accuracy: 91.61% |                                           |    |    |    |    |    |     |    |    |    |     |

Analysis of GFCC(39) Features  
with KNN Algorithm



Accuracy of GFCC (39) Features  
with KNN Algorithm

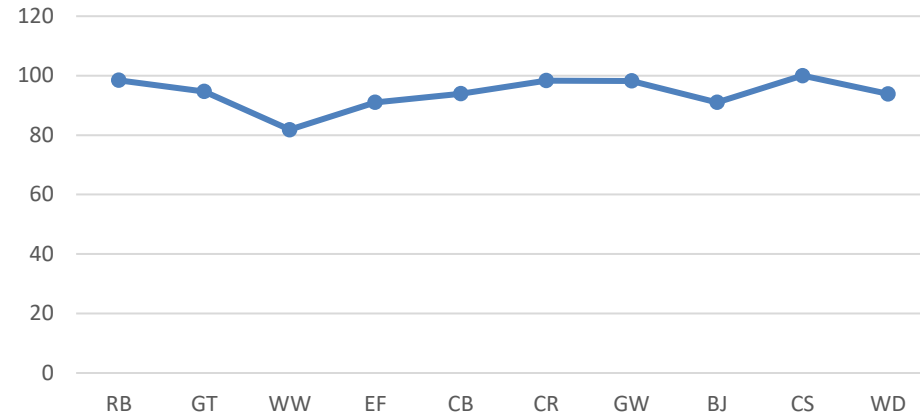
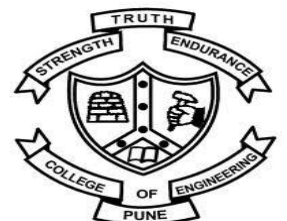


Fig 27 : Analysis and Accuracy of GFCC (39) Features with KNN Algorithm

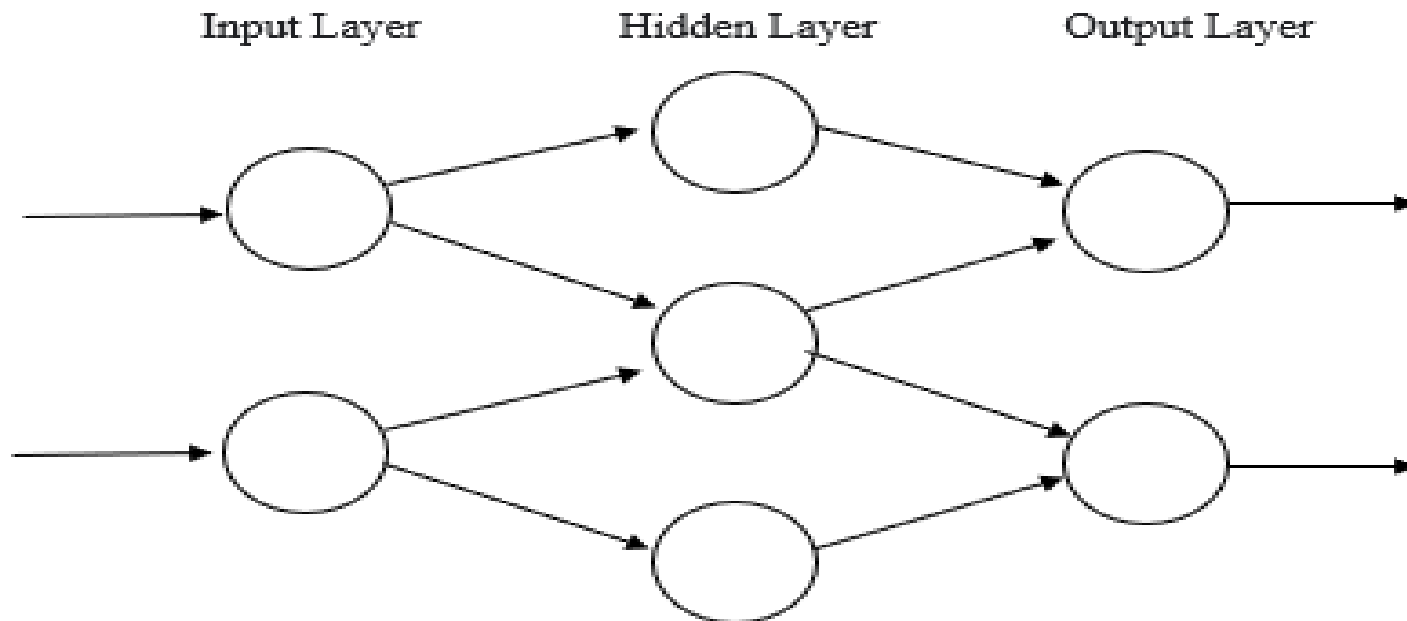
Table 28:Overall Accuracy of Feature Extraction using KNN algorithm

| <b>Sr No.</b> | <b>Feature vector</b> | <b>MFCC</b> | <b>LFCC</b> | <b>GFCC</b> |
|---------------|-----------------------|-------------|-------------|-------------|
| 1             | 13 coefficients       | 91.13       | 91.8        | 91.64       |
| 2             | 26 coefficients       | 91.5        | 91.76       | 92.32       |
| 3             | 39 coefficients       | 93.64       | 91.9        | 91.61       |



# Artificial Neural Network

- ANN contains three layers called the Input layer, hidden layer and output layer.
- The ReLU (Rectified Linear Unit) activation function assigns value zero if the negative and same value is positive.



Figure\_28: Block Diagram of ANN Model

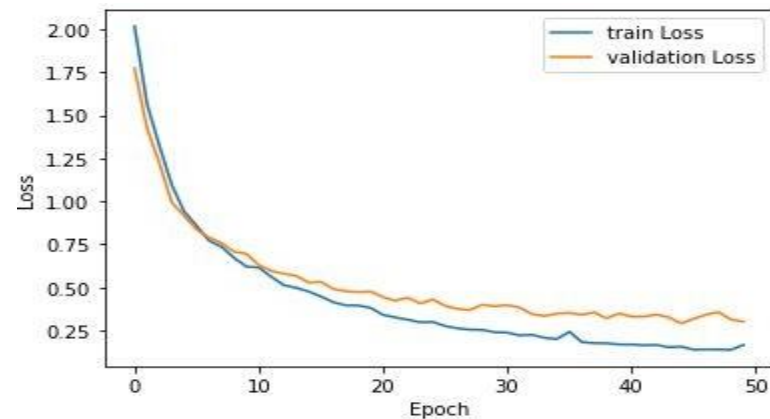
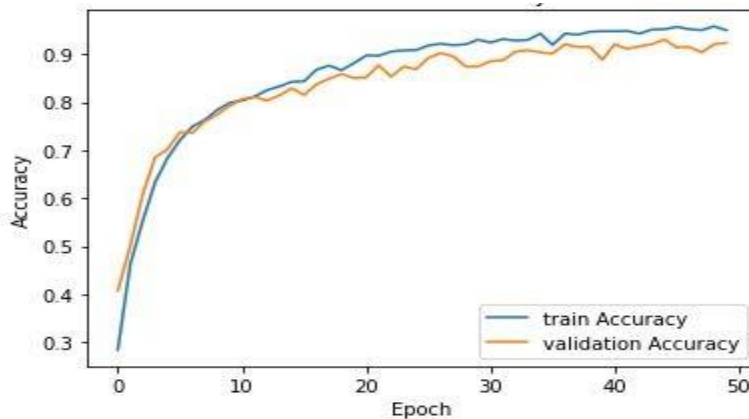


Figure 29: Training and Validation Accuracy and loss of 13 MFCC coefficients

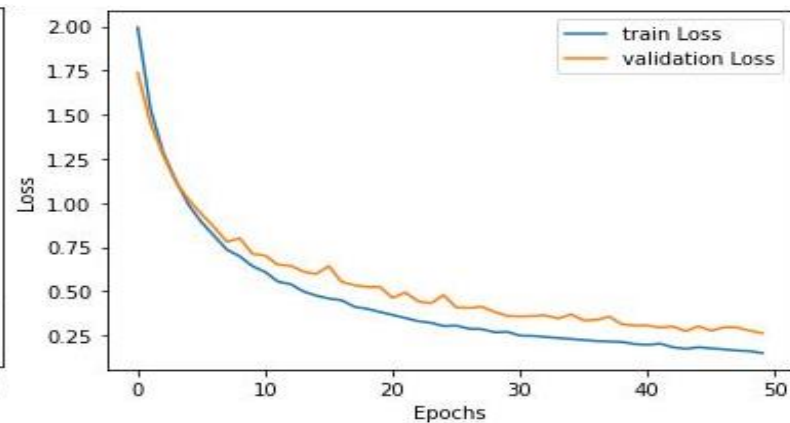
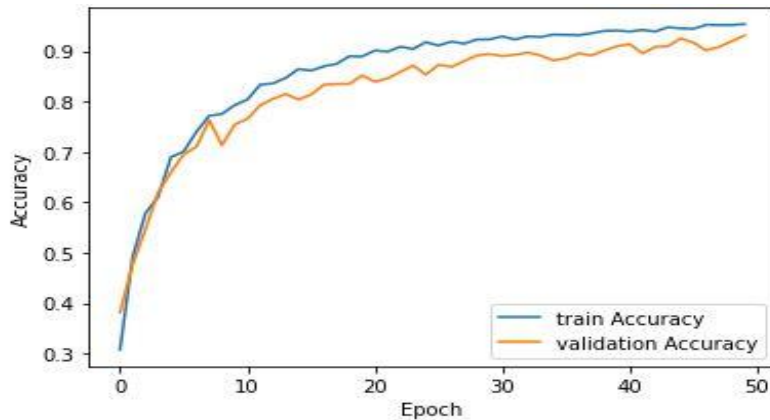


Figure 30: Training and Validation Accuracy and loss of 26 MFCC coefficients

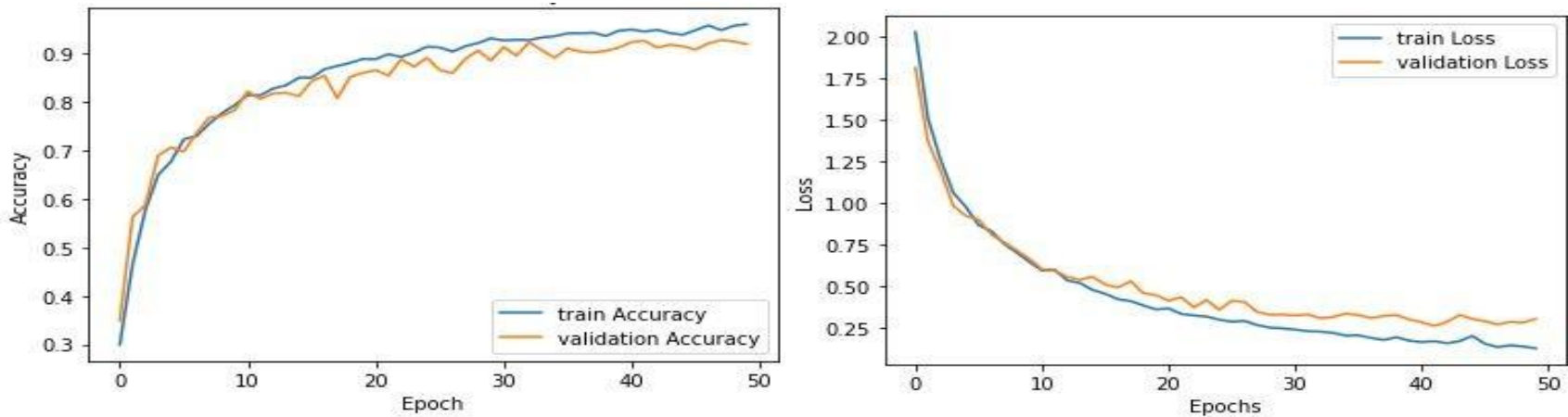


Figure 31: Training and Validation Accuracy and loss of 39 MFCC coefficients

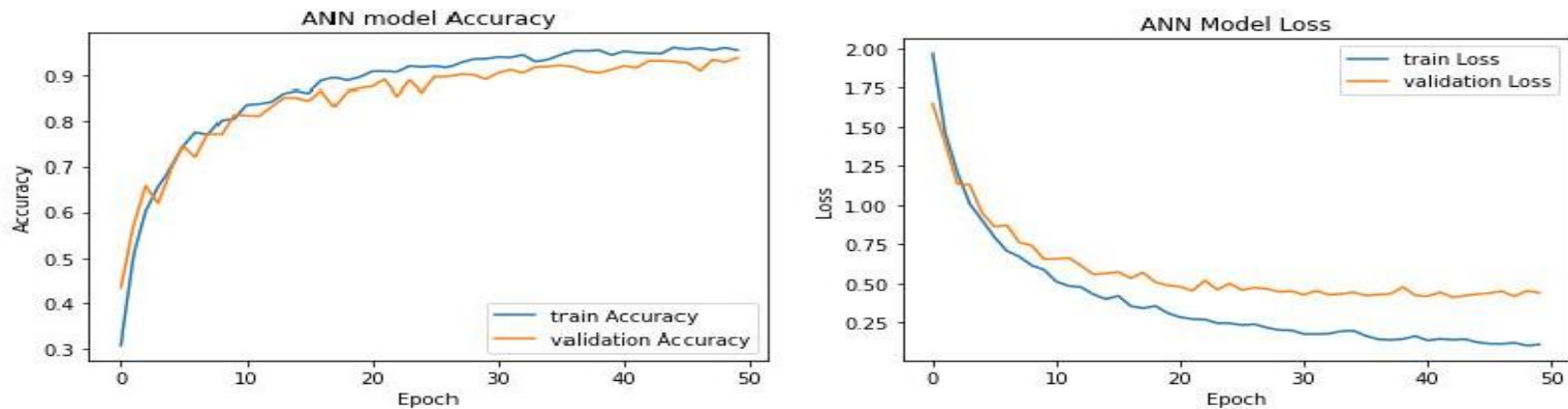


Figure 32: Training and Validation Accuracy and loss of 27 MFCC coefficients



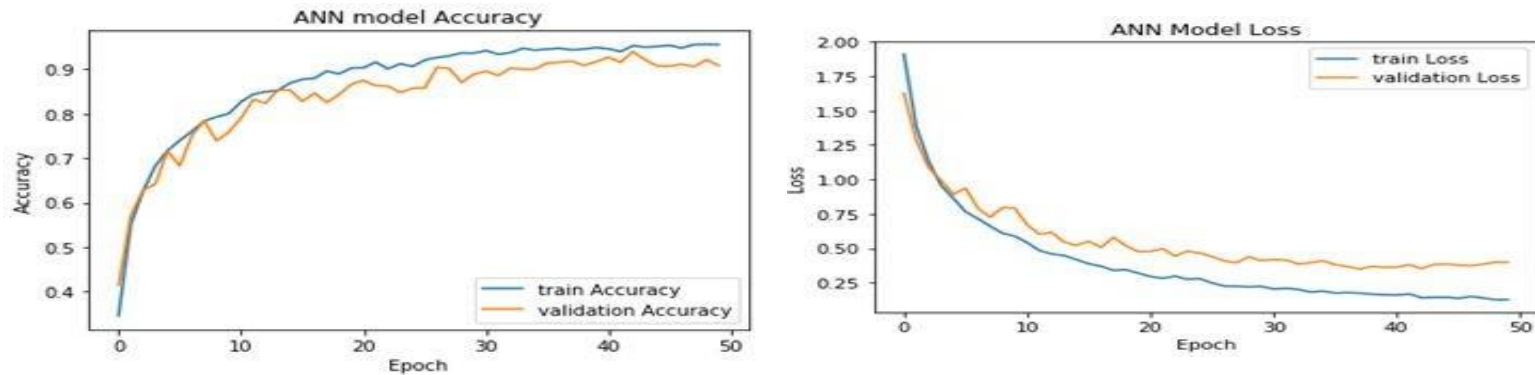


Figure 33: Training and Validation Accuracy and loss of 13 LFCC coefficients

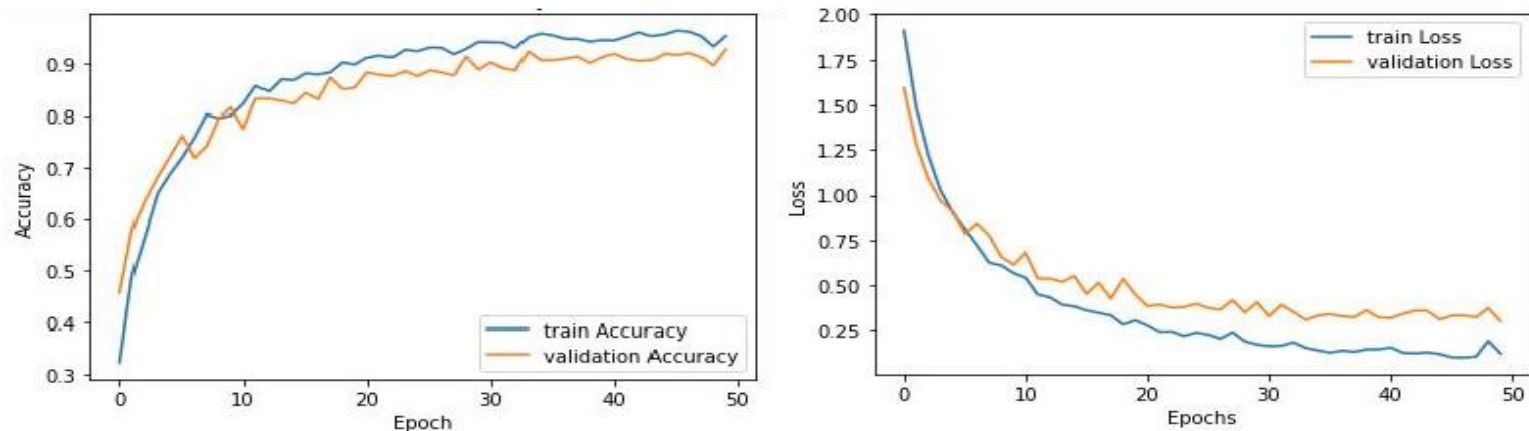


Figure 34: Training and Validation Accuracy and loss of 26 LFCC coefficients

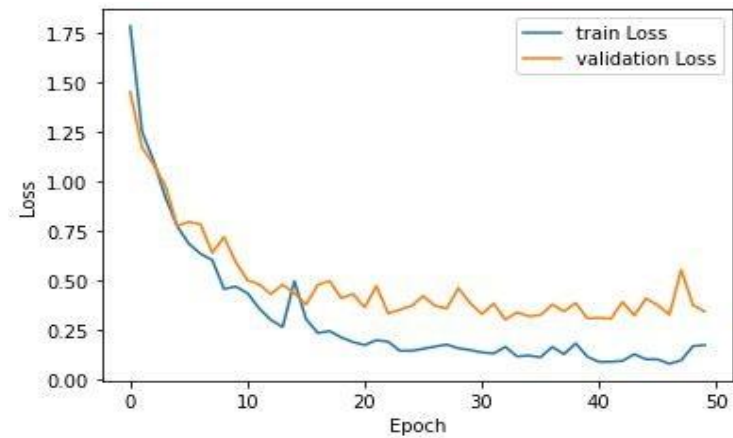
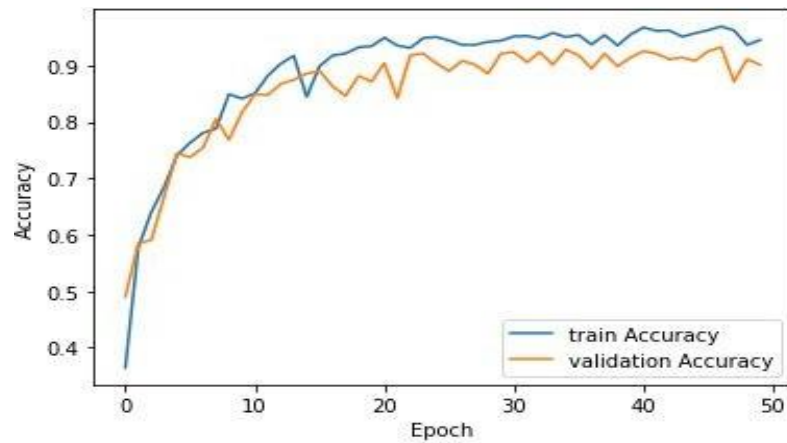


Figure 35: Training and Validation Accuracy and loss of 39 LFCC coefficients

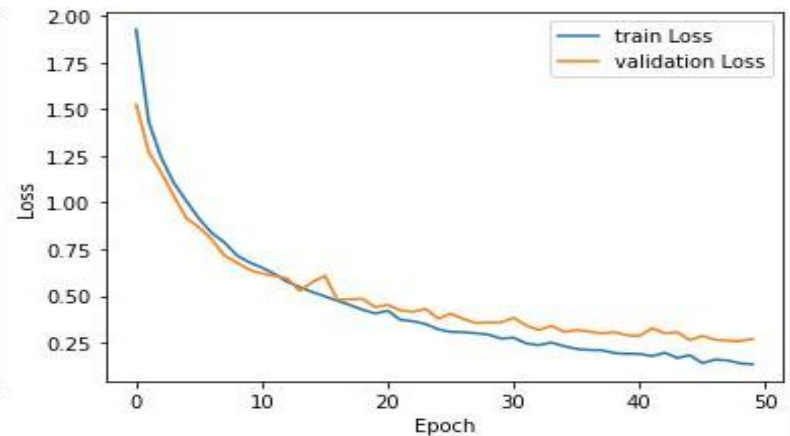
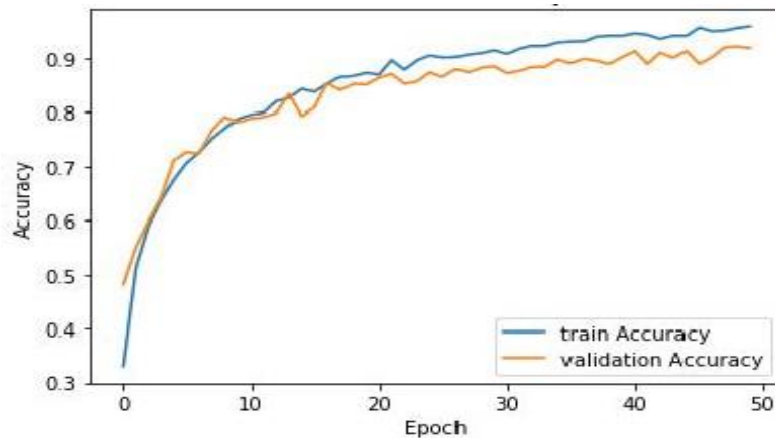


Figure 36: Training and Validation Accuracy and loss of 27 LFCC coefficients

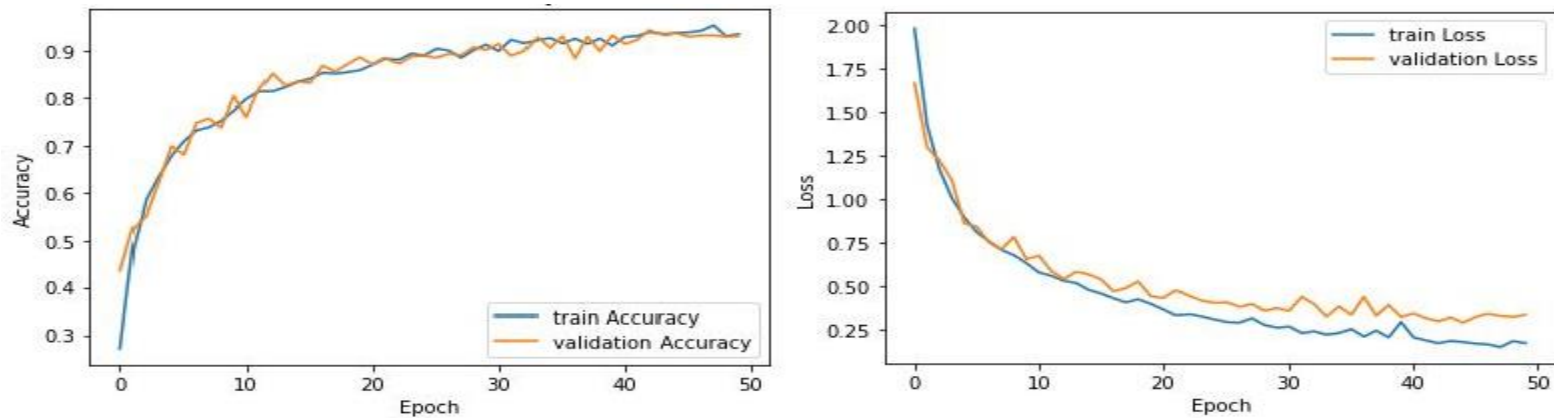


Figure 37: Training and Validation Accuracy and loss of 13 GFCC coefficients

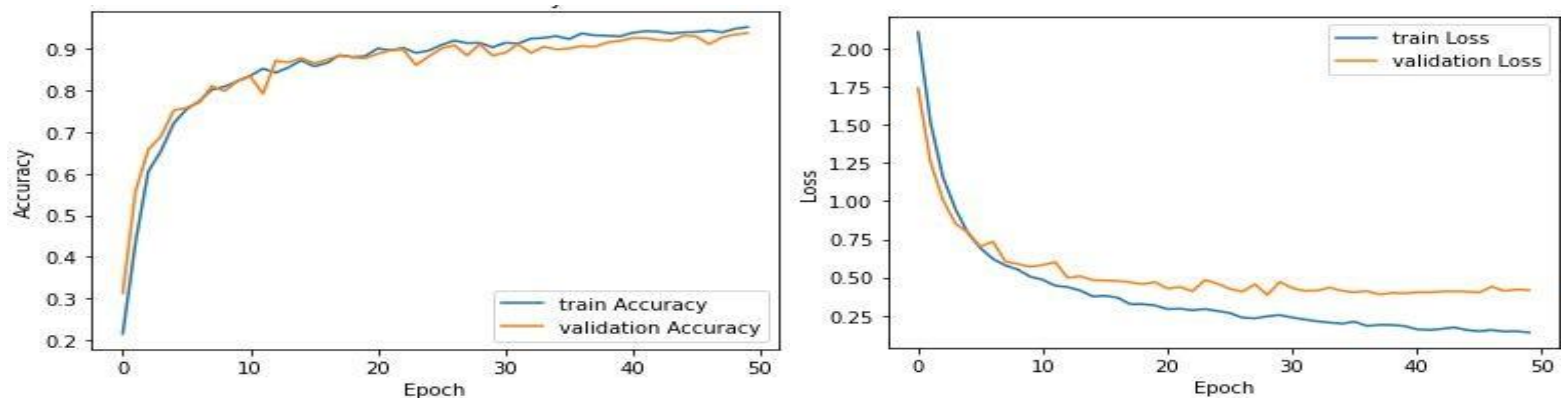


Figure 38: Training and Validation Accuracy and loss of 26 GFCC coefficients

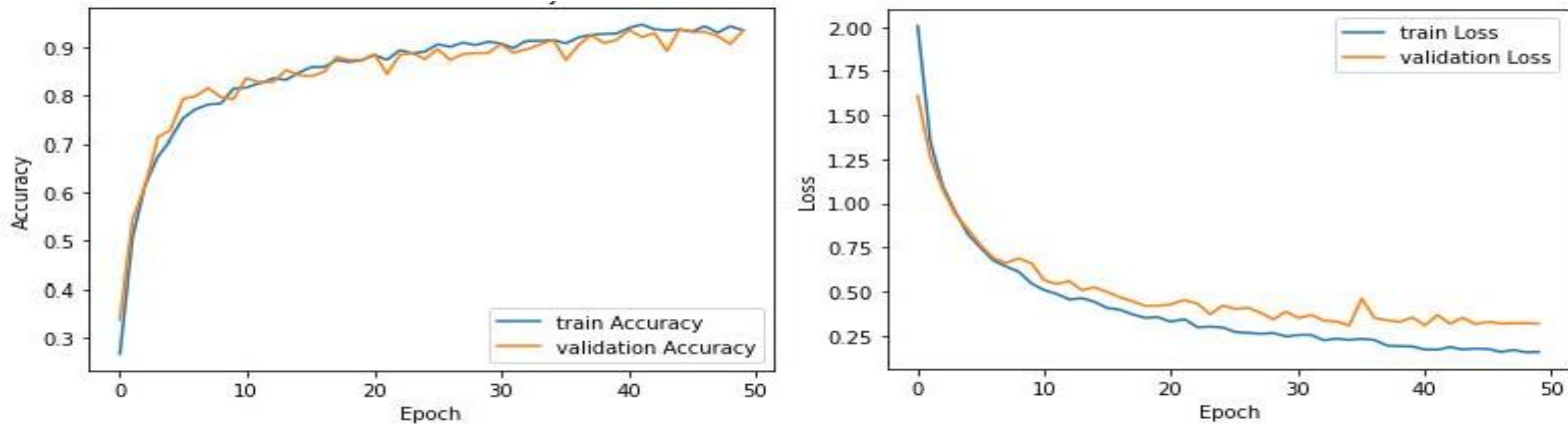


Figure 39: Training and Validation Accuracy and loss of 39 GFCC coefficients

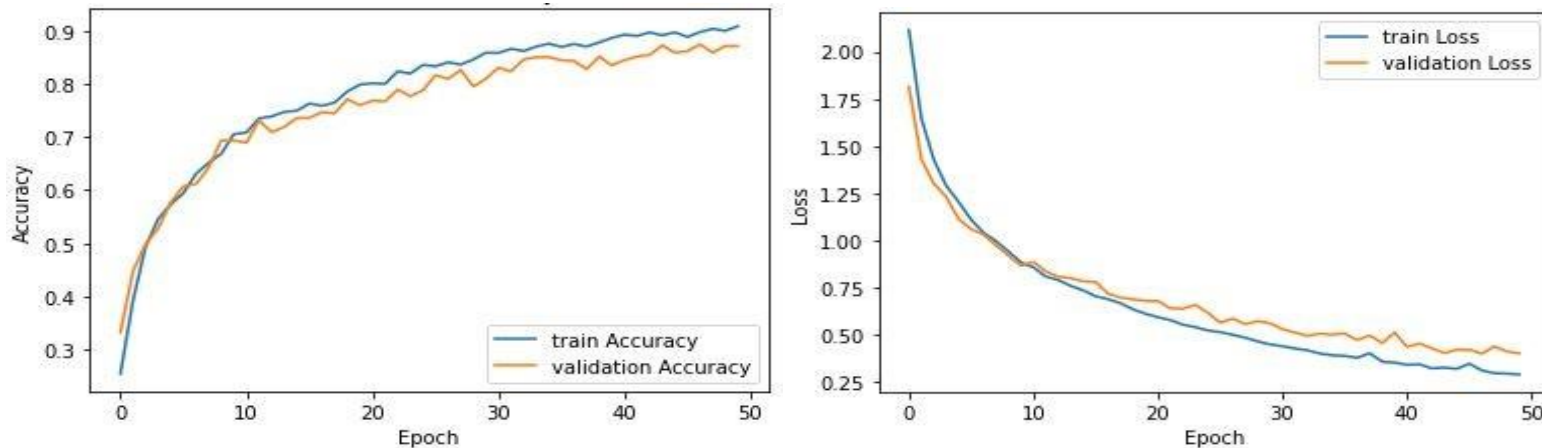
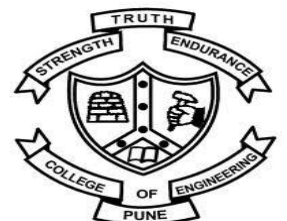


Figure 40: Training and Validation Accuracy and loss of 27 GFCC coefficients

Table 29:Overall Accuracy of Feature Extraction using ANN algorithm

| Sr No. | Feature vector  | MFCC  | LFCC  | GFCC  |
|--------|-----------------|-------|-------|-------|
| 1      | 13 coefficients | 91.6  | 92.03 | 92.67 |
| 2      | 26 coefficients | 92.07 | 91.66 | 92.87 |
| 3      | 39 coefficients | 91.94 | 91.88 | 92.33 |
| 4      | 27 coefficients | 94.7  | 92.6  | 87.88 |



# Convolutional Neural Network

- The layers of CNN contain the Input layer, a hidden layer and an output layer.
- It also includes multiple convolution layers, pooling layers. The convolution network is a 2-D convolution layer.

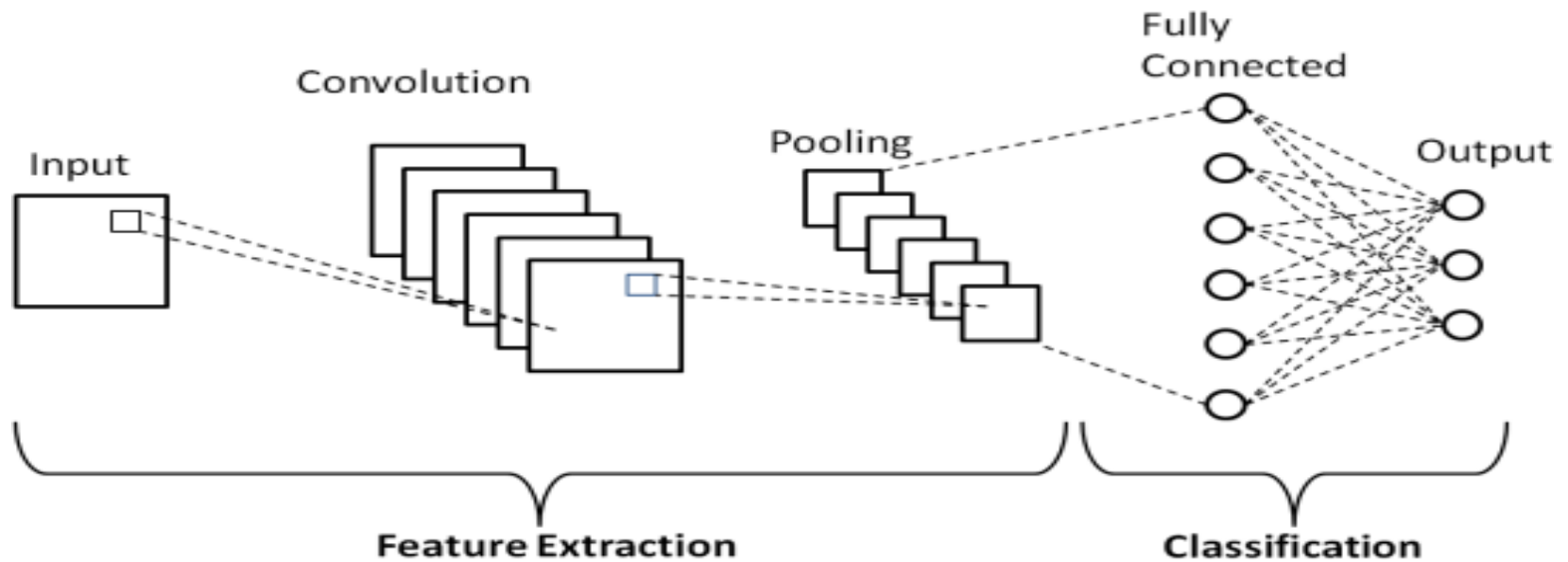


Figure 41:Block Diagram of CNN Architecture

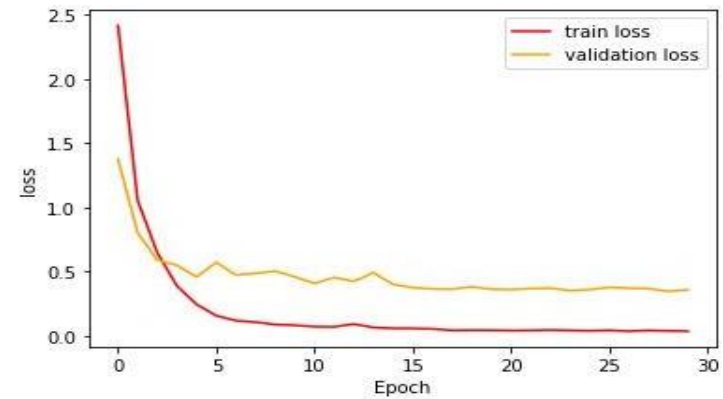
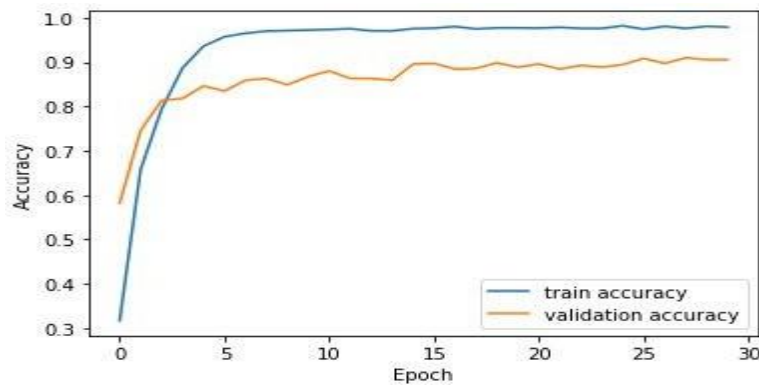


Figure 42: Training and Validation Accuracy and loss of 13 MFCC coefficients

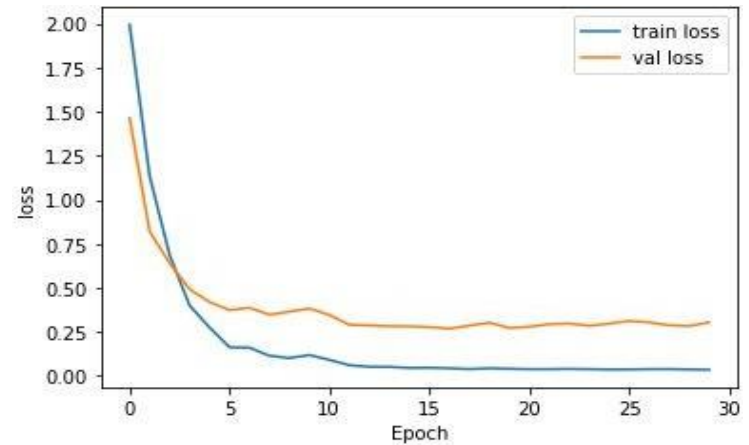
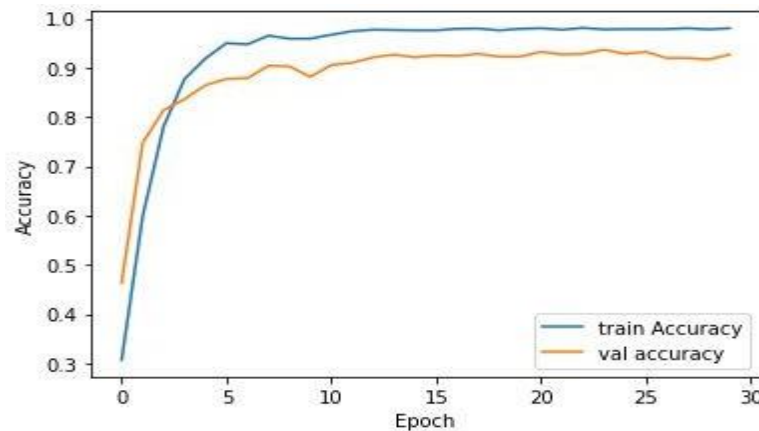


Figure 43: Training and Validation Accuracy and loss of 26 MFCC coefficients



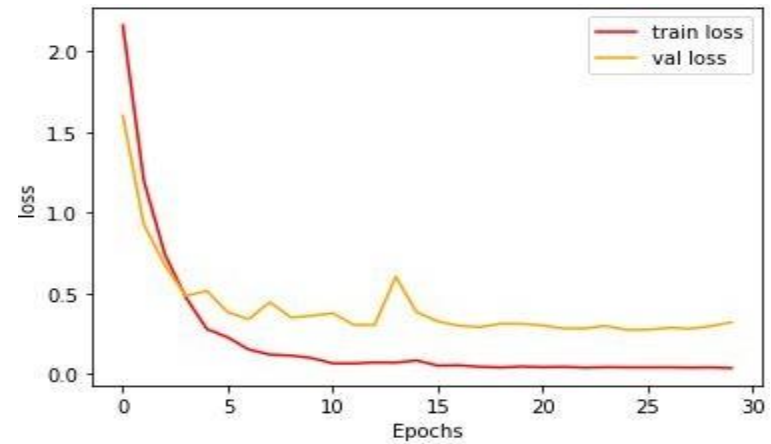
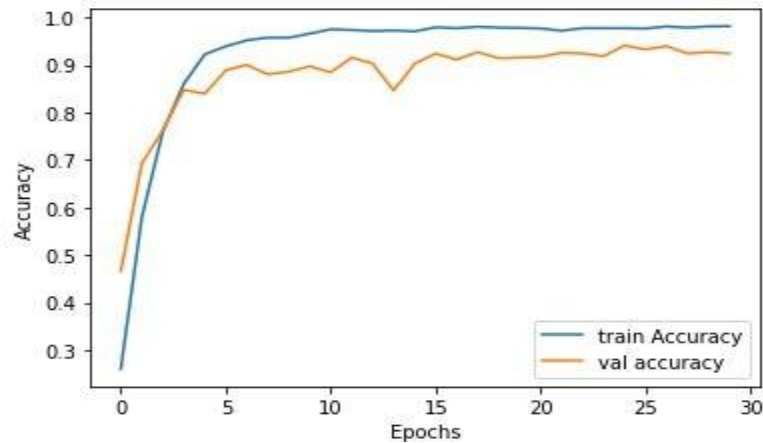


Figure 44: Training and Validation Accuracy and loss of 39 MFCC coefficients

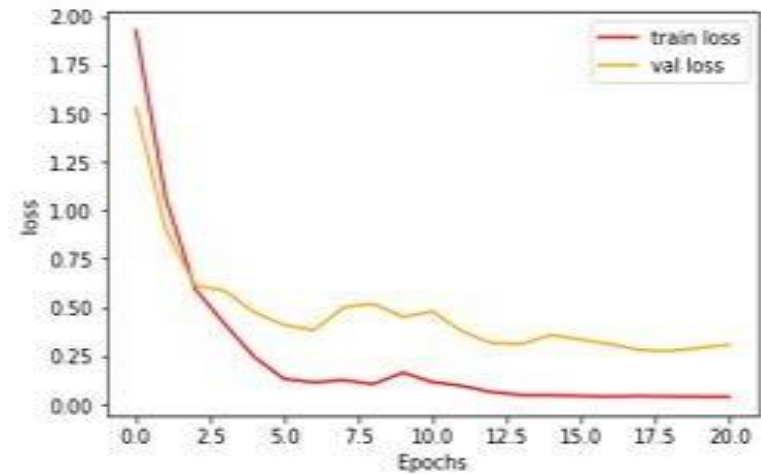
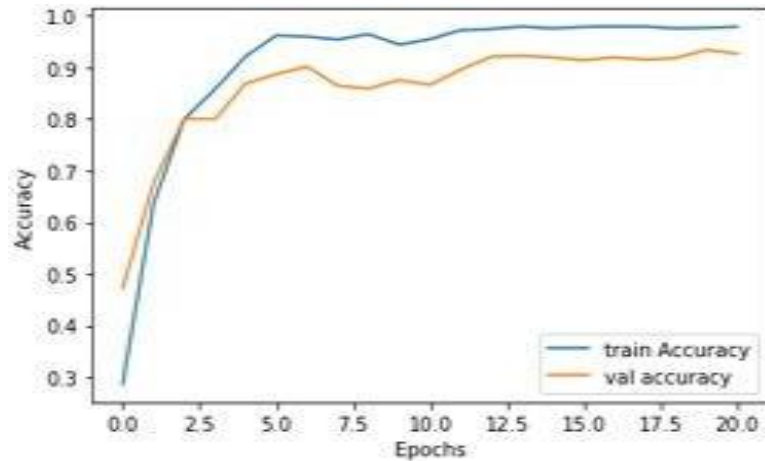


Figure 45: Training and Validation Accuracy and loss of 27 MFCC coefficients



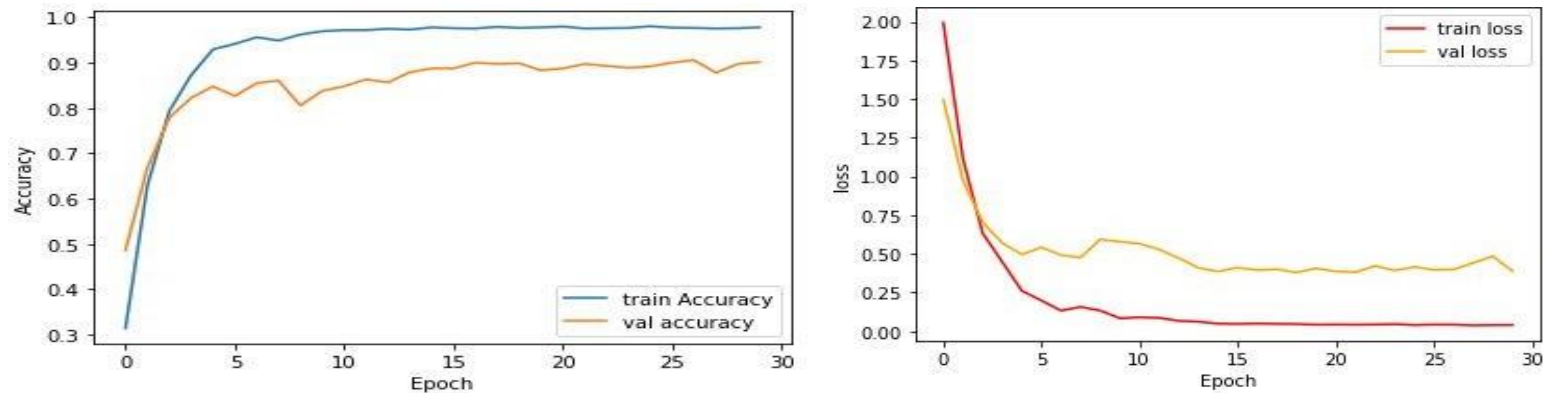


Figure 46: Training and Validation Accuracy and loss of 13 LFCC coefficients

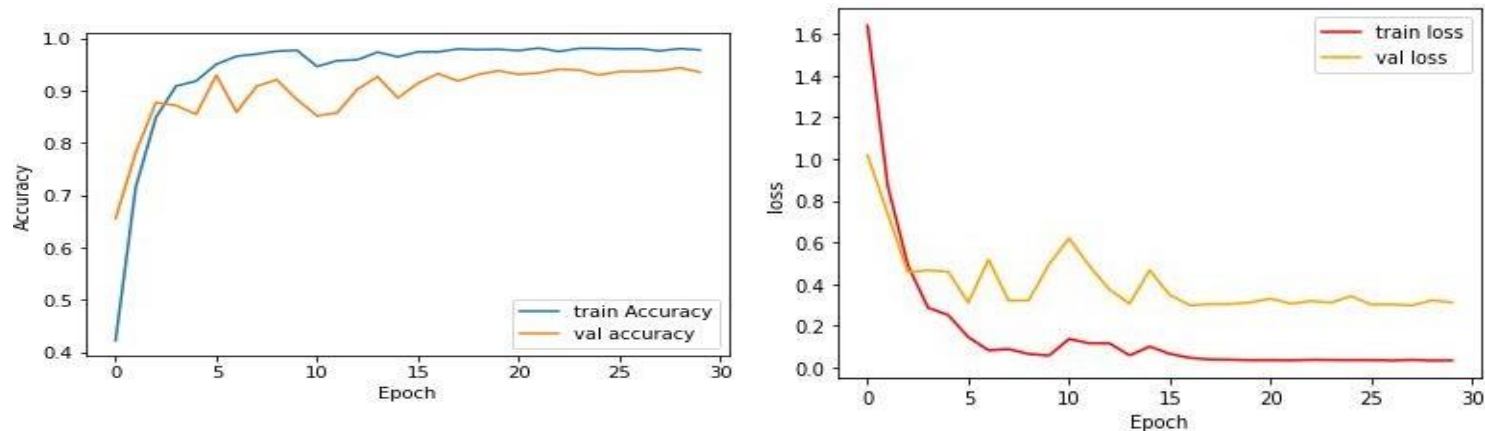


Figure 47: Training and Validation Accuracy and loss of 26 LFCC coefficients

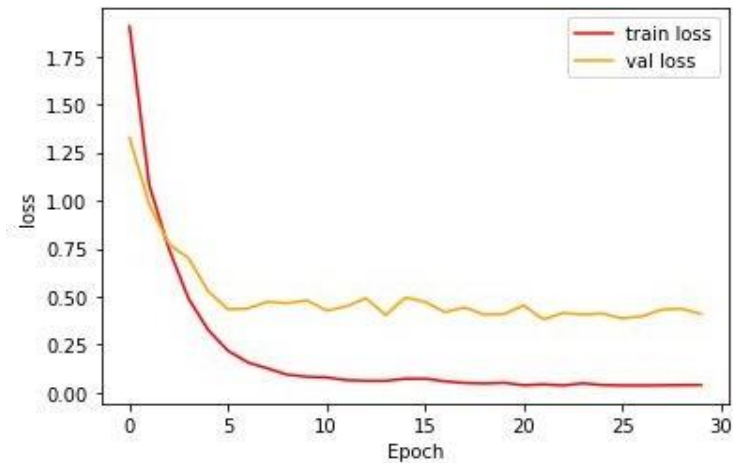
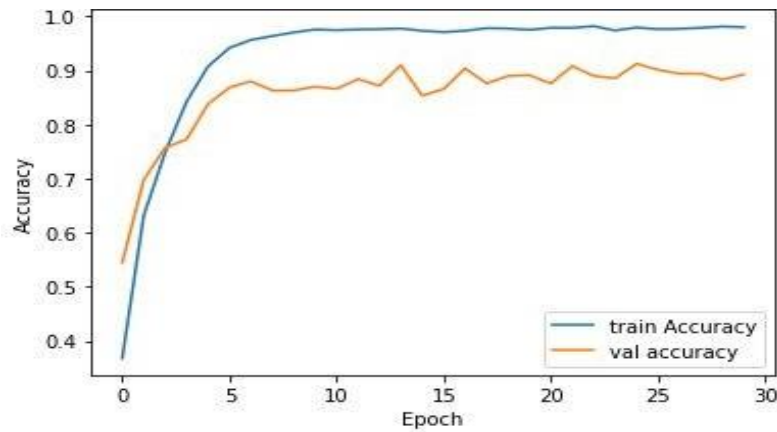


Figure 48: Training and Validation Accuracy and loss of 39 LFCC coefficients

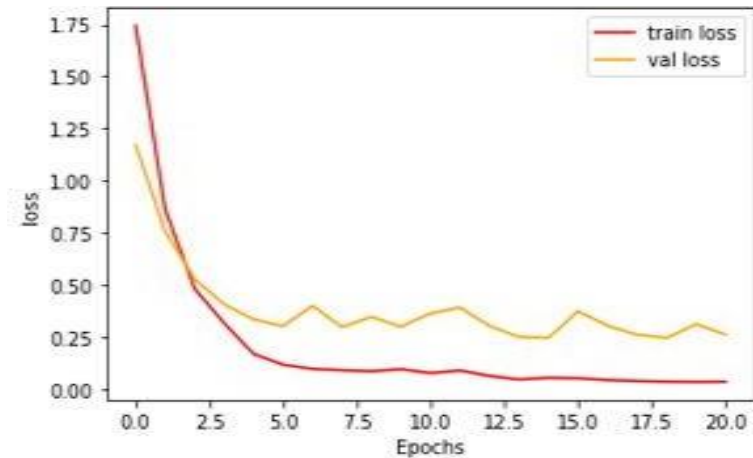
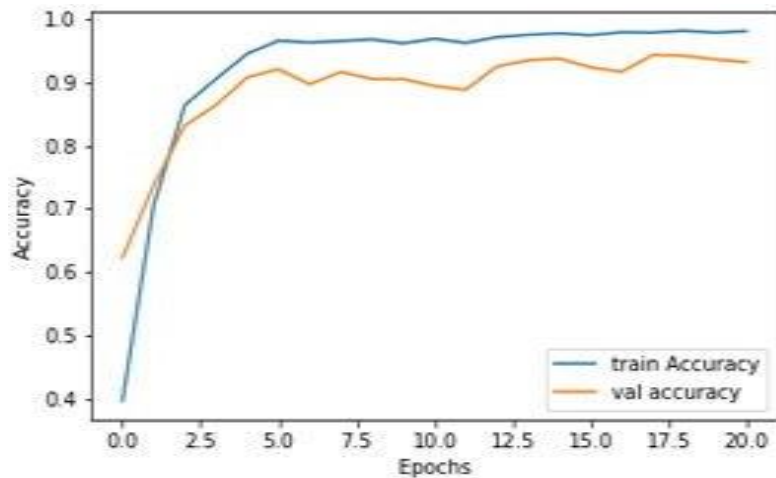


Figure 49: Training and Validation Accuracy and loss of 27 LFCC coefficients

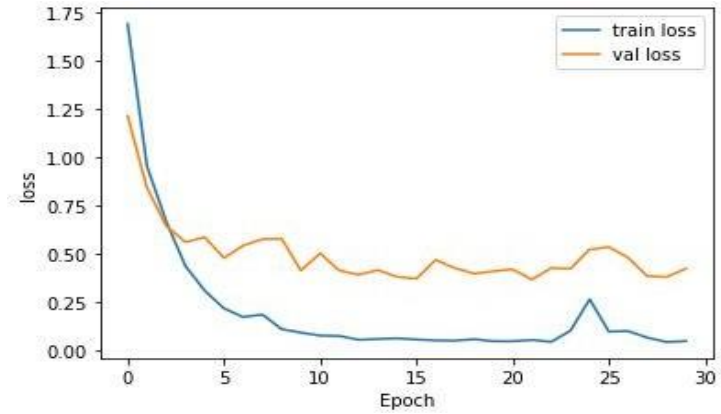
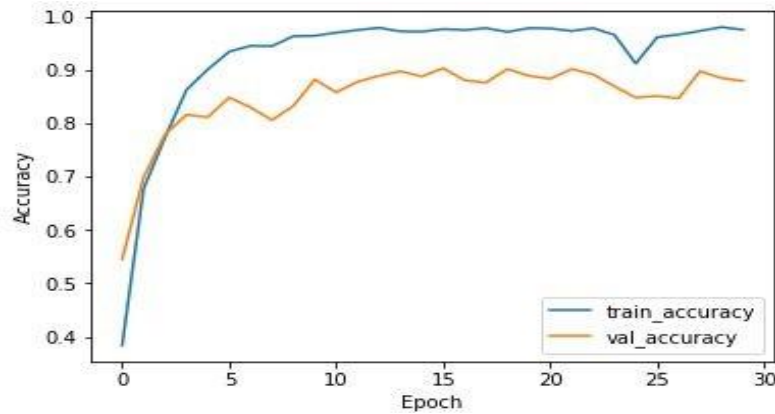


Figure 50: Training and Validation Accuracy and loss of 13 GFCC coefficients

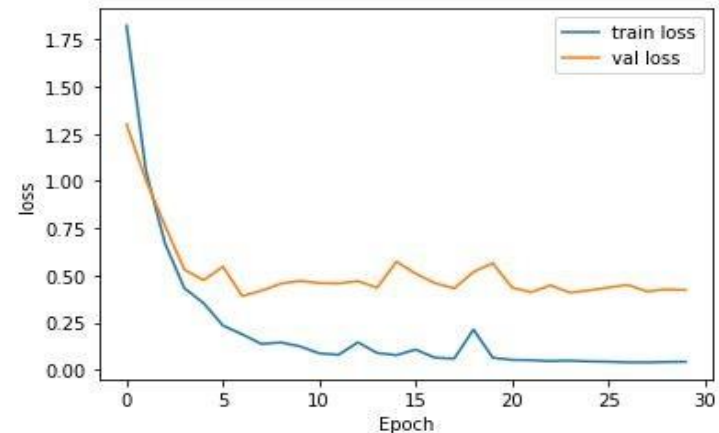
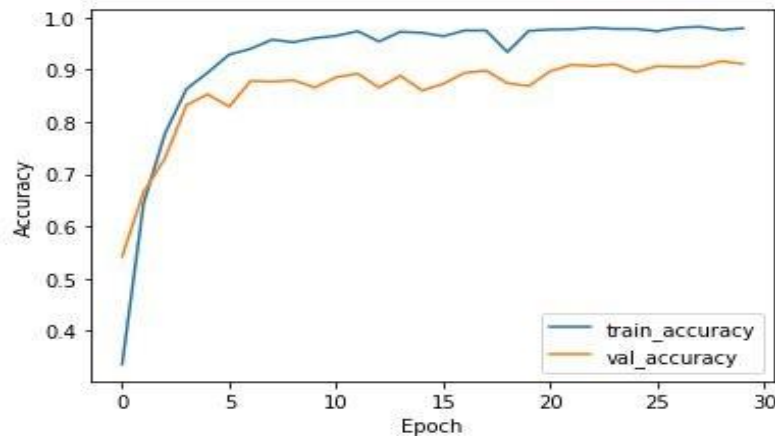


Figure 51: Training and Validation Accuracy and loss of 26 GFCC coefficients

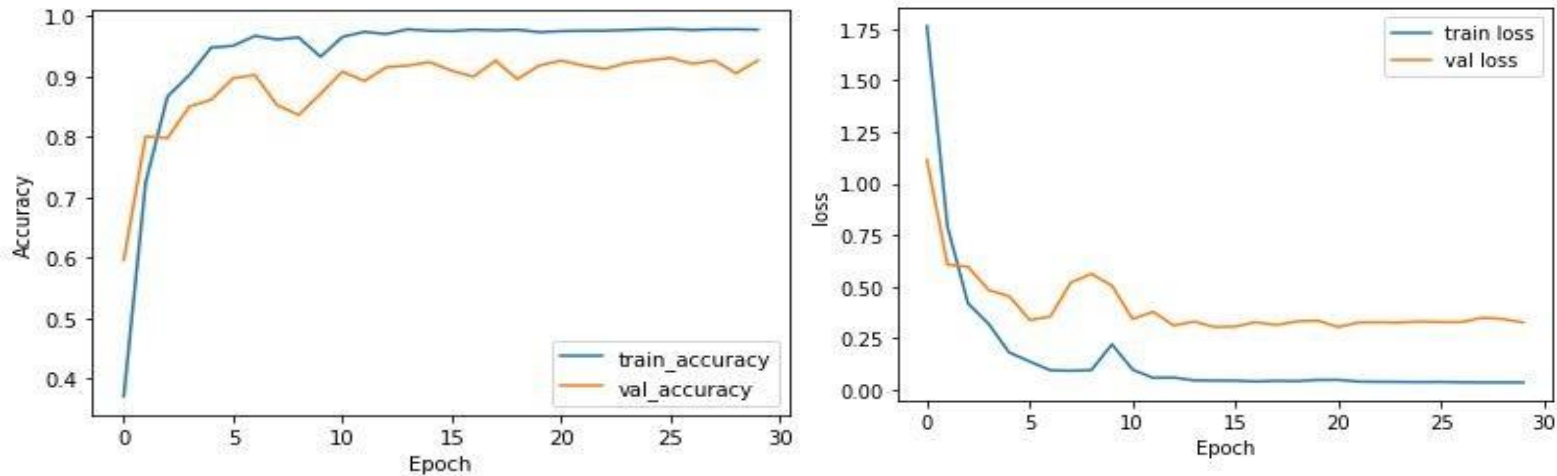


Figure 52: Training and Validation Accuracy and loss of 39 GFCC coefficients

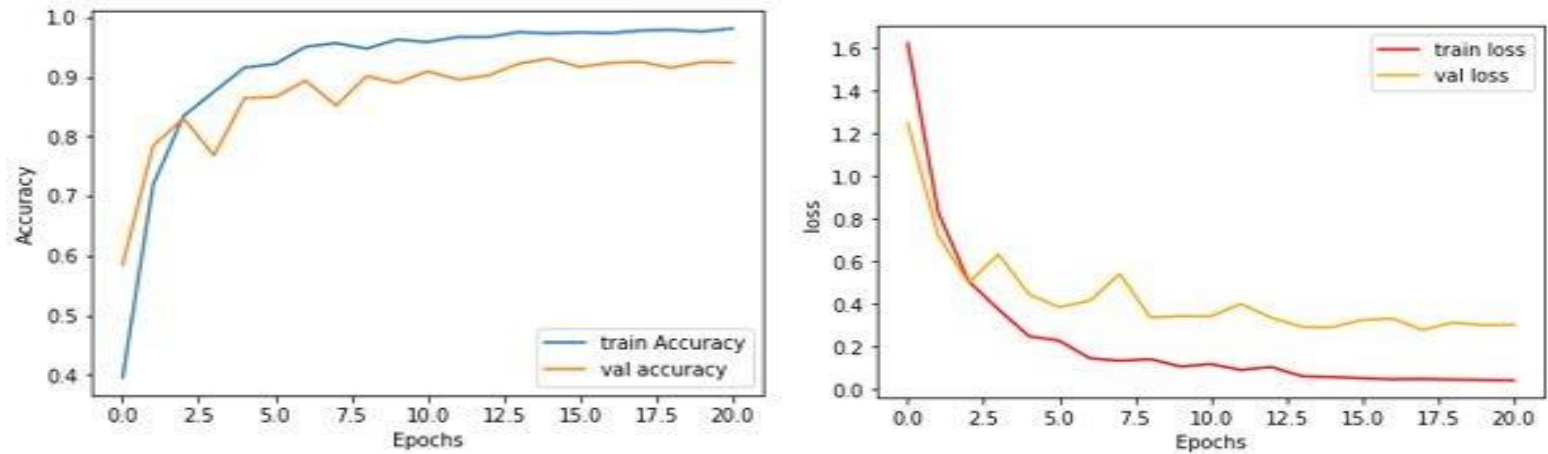


Figure 53: Training and Validation Accuracy and loss of 27 GFCC coefficients

Table 30:Overall Accuracy of Feature Extraction using CNN algorithm

| <b>Sr No.</b> | <b>Feature vector</b> | <b>MFCC</b> | <b>LFCC</b> | <b>GFCC</b> |
|---------------|-----------------------|-------------|-------------|-------------|
| 1             | 13 coefficients       | 91.6        | 92.03       | 92.67       |
| 2             | 26 coefficients       | 92.07       | 91.66       | 92.87       |
| 3             | 39 coefficients       | 91.94       | 91.88       | 92.33       |
| 4             | 27 coefficients       | 94.7        | 92.6        | 87.88       |

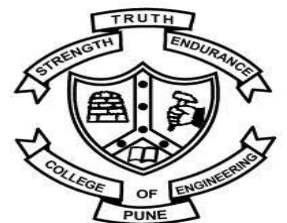
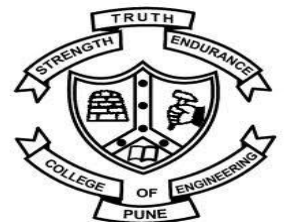


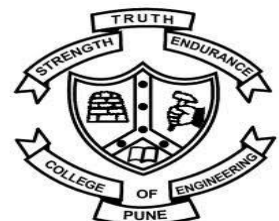
Table 31: Classification accuracy of Different Feature set and classifiers

| Sr.No | Feature Set                                        | Length of<br>Feature Vector | Accuracy |        |        |        |
|-------|----------------------------------------------------|-----------------------------|----------|--------|--------|--------|
|       |                                                    |                             | KNN      | SVM    | ANN    | CNN    |
| 1     | Mel Frequency<br>Cepstral<br>Coefficients          | 13                          | 91.13%   | 82.38% | 91.60% | 91.13% |
|       |                                                    | 26                          | 91.50%   | 81.33% | 92.07% | 90.36% |
|       |                                                    | 39                          | 93.64%   | 82.21% | 91.94% | 92.70% |
|       |                                                    | 27                          | 94.70%   | 73.58% | 94.70% | 93.60% |
| 2     | Linear Frequency<br>Cepstral<br>Coefficients       | 13                          | 91.80%   | 85.53% | 92.03% | 91.67% |
|       |                                                    | 26                          | 91.76%   | 85.49% | 91.66% | 90.67% |
|       |                                                    | 39                          | 91.90%   | 85.88% | 91.88% | 92.80% |
|       |                                                    | 27                          | 95.05%   | 87.56% | 92.60% | 93.80% |
| 3     | Gammatone<br>Frequency<br>Cepstral<br>coefficients | 13                          | 91.64%   | 87.64% | 92.67% | 91.58% |
|       |                                                    | 26                          | 92.32%   | 87.44% | 92.87% | 91.18% |
|       |                                                    | 39                          | 91.61%   | 85.88% | 92.33% | 90.18% |
|       |                                                    | 27                          | 91.66%   | 85.22% | 87.88% | 93.44% |



# Conclusion

- Three sets of classifiers and feature sets are selected to classify bird species.
- Gammatone Frequency Cepstral Coefficients with feature selection technique gives an accuracy of 93.44 %.
- Linear Frequency Cepstral Coefficients with feature selection techniques offers maximum accuracy of 95.05 %.
- The Recursive Feature Elimination (RFE) algorithm further improves the accuracy.
- CNN with MFCC as features gives the highest accuracy compared to LFCC and GFCC.

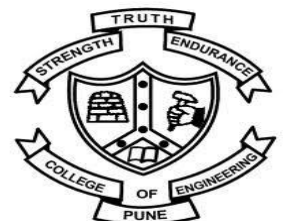


# Research Paper

|                               |                                                                                       |
|-------------------------------|---------------------------------------------------------------------------------------|
| Conference Name               | 8th International Conference on Signal Processing and Integrated Networks (SPIN 2021) |
| Paper Title                   | Bird Classification Based on Bird Sounds                                              |
| Status                        | ACCEPTED                                                                              |
| Acceptance Notification       | 13-07-2021                                                                            |
| Conference Date               | <b>26-27 August 2021</b>                                                              |
| IEEE Conference Record Number | 52536                                                                                 |

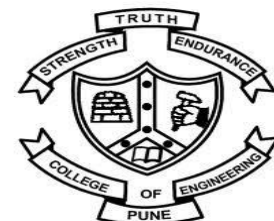
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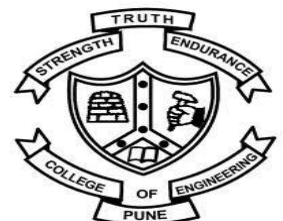




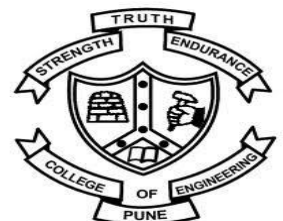
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# Thank You

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